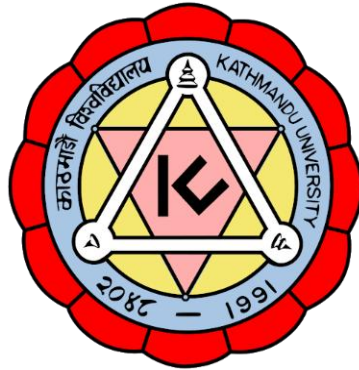


KATHMANDU UNIVERSITY
SCHOOL OF ENGINEERING
DEPARTMENT OF CIVIL ENGINEERING



PROJECT REPORT ON
“ESTIMATION AND VALUATION OF RESIDENTIAL HOUSE”

SUBMITTED TO:

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ABSTRACT

The purpose of this project was to estimate the value of my own residential building located in Dhulikhel, Nepal. The project involved conducting a thorough analysis of the property, including its location, design, and construction materials. Upon Architectural and Structural drawings of building, the detailed estimation was carried out. This information was used to estimate the value of the subject property using Districts Rates. The results of the estimation and valuation were presented in a report. The project provided a comprehensive assessment of the value of the subject property.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Er. Raju Bhele for providing me with the opportunity to work on the project titled "Estimation and Valuation of own residential Building." His guidance and support throughout the project were invaluable and greatly contributed to the success of the project. I am grateful for his timely feedback and advice, which helped me to improve the quality of the project report. I would also like to thank him for his patience and encouragement, which helped me to overcome any challenges that I faced during the project. I am grateful for the support and opportunities that he has provided me and I am confident that this project will be a valuable addition to my portfolio.

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A. OBJECTIVES, SCOPE AND LIMITATIONS

1. OBJECTIVES OF THE PROJECT

The objectives of the project are as follows:

- a. To estimate the value of a residential building located in Dhulikhel, Nepal.
- b. To conduct a thorough analysis of the property, including its location, design, and construction materials.
- c. To gather data on similar properties in the area and their sale prices through market research.
- d. To learn to value an existing residential building.

2. SCOPE OF THE PROJECT

The main aim of the project is to prepare the detailed drawing of a house in AutoCAD and estimate the present construction cost for the building based on the prepared detailed architectural and structural drawings. The scope of this project includes:

- a. Preparation of detailed architectural and structural drawings of a building in AutoCAD.
- b. Use of NBC 201 and NBC 205 (Mandatory rule of thumb for reinforced concrete buildings).

3. LIMITATIONS OF THE PROJECT

The limitations of the project are as follows:

- a. CAD Drawings are incomplete of various items such as reserve tank, septic tank and so on.
- b. The project report will not include a detailed analysis of the property's financial performance or the potential for future depreciation in value.
- c. Estimation of various items work has not been carried out. They include Chipping, Screeding and so on.
- d. Structural drawing of Reserve tank, water flow, sanitary etc. have not been incorporated.

B. INTRODUCTION

ESTIMATION:

Estimation refers to the process of determining the cost, quantity, and other measures of a construction project. Estimation involves making educated guesses or predictions based on incomplete or limited information. It is a critical part of the planning and design process for any construction project, as it helps to determine the feasibility of a project and provides a basis for budgeting and scheduling.

There are many factors that must be considered when making estimates in civil engineering, including the materials and labor required, the complexity of the project, and any potential risks or challenges that may arise. Estimators use a variety of tools and techniques, such as cost databases, computer software, and historical data, to help them make accurate predictions. Estimation is an important skill for civil engineers, as it allows them to plan and manage projects effectively and ensure that they are completed within budget and on schedule.

METHOD OF ESTIMATION

- i. Estimate: An estimate is a computation or calculation of quantities required in the construction of the work
- ii. Actual cost: The actual cost of work is known at the completion of work. In engineering project, we prepare an estimated cost of the project beforehand, which includes the cost of material, labors, etc. If the estimated cost is near to the actual cost, it is said to be good estimation.
- iii. Detailed estimate: Preparation of detailed estimated consists of working out of quantities.

Table 1 : Details of Measurement

S.N.	Item	No	Length	Breadth	Height	Quantity	Unit
1	Earthwork						
2	Concrete						
3	Brickwork						

- iv. Abstract Cost: The cost under item of work is calculated from the quantities already calculated.

Table 2 : Abstract of Estimate

S.N.	Item	Quantity	Unit of Quantity	Rate	Unit of Rate	Amount	Remarks
1	Earthwork						
2	Concrete						

3	Brickwork						
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MAJOR ITEMS OF WORK:

1. SITE CLEANING:

Site cleaning is an important aspect of civil engineering that involves the removal of unwanted stuffs and other materials from a construction site. Site cleaning is typically carried out by a team of workers using specialized equipment, such as excavators, bulldozers, and dump trucks. The process may also involve the use of tools such as shovels, brooms, and other hand tools to remove smaller debris and clean up the site. Site cleaning is an important step in the construction process, as it helps to ensure that the site is safe and ready for the next phase of development.

2. EARTHWORK IN EXCAVATION:

Earthwork in excavation refers to the process of moving and shaping the earth to create a desired grade or slope. This is a common aspect of construction projects, particularly in the building of roads, bridges, and other infrastructure projects. Earthwork typically involves the removal of excess soil or rock from a site, as well as the addition of fill material to create a stable and level surface. Excavation may be carried out using a variety of equipment, such as bulldozers, excavators, and backhoes, depending on the size and complexity of the project.

Earthwork in excavation is an important aspect of many construction projects, as it helps to create a stable and level foundation for buildings, roads, and other structures. It requires careful planning and attention to detail to ensure that the work is completed safely and to the required specifications.



Figure 1: Earthwork Excavation

3. BRICK SOILING:

The bricks to be used for soling should be dry and placed on the bed with the frog facing upwards. The base upon which the brick soling is to be laid should be leveled and compacted properly. The bricks should be placed accurately in line, leveled and arranged as required. The bricks should be tightly packed together as much as possible and any gaps should be filled with sand. It is not allowed to use cut bricks for soling unless specifically stated. After laying the bricks for soling, they should be compacted to ensure a uniform and tight finish without damaging the felt underneath. The finished surface should be level in both the



longitudinal and transverse directions.

Figure 2 : Brick Soiling

<https://allaboutcivil.org/what-is-soling-why-it-is-done-types-procedure/>

4. EARTH FILLING:

Earth filling refers to the process of adding soil or other fill material to a site in order to create a stable and level surface. This is a common aspect of construction projects, particularly in the building of roads, bridges, and other infrastructure projects. Earth filling typically involves the removal of excess soil or rock from a site, as well as the addition of fill material to create a stable and level surface. The fill material may be sourced from on-site or off-site, depending on the availability and cost of the material. It helps to create a stable and level foundation for buildings, roads, and other structures. It requires careful planning and attention to detail to ensure that the work is completed safely and to the required specifications.

5. REINFORCEMENTS:

Reinforcements are materials used to strengthen and support structures, such as concrete, masonry, and wood. They are used in a variety of construction applications, including buildings, bridges, and other infrastructure projects. Reinforcements are an important aspect



of construction, as they help to strengthen and support structures and ensure their stability and durability over time. They are used in a variety of applications and can be made from a range of materials, depending on the specific needs of the project.

Figure 3 : RCC Work

<https://www.walcoom.com/products/architecturalmesh/construction-materials/gallery.html>

6. CEMENT CONCRETE WORK:

Cement concrete is a construction material made from a mixture of cement, water, sand, and aggregate (such as crushed stone or gravel). It is widely used in a variety of construction projects, including buildings, roads, bridges, and other infrastructure projects. Cement concrete works involve the use of specialized equipment and techniques, such as mixers, concrete pumps, and finishing tools, to create a wide range of structures and surfaces. It requires careful planning and attention to detail to ensure that the work is completed safely and to the required specifications.

7. FORMWORK:

Formwork refers to the temporary structures that are used to support and shape concrete or other materials during the construction process. Formwork is an important aspect of many construction projects, as it helps to ensure that the finished product meets the required specifications and tolerances. It helps to ensure that the finished product meets the required specifications and tolerances. It is used in a variety of applications and can be made from a



range of materials, depending on the specific needs of the project.

Figure 4 : Formwork

8. BRICK MASONRY WORK:

Brick masonry is a construction technique that involves the use of bricks and mortar to create walls, buildings, and other structures. Brick masonry is a traditional and widely-used method of construction that has been in use for centuries. The process of brick masonry involves laying bricks in a pattern using a wet, sticky mixture of cement, sand, and water known as mortar. The bricks are carefully aligned and spaced to create a strong and durable structure. Brick masonry is a skilled trade that requires knowledge of materials, tools, and techniques. It is an important aspect of many construction projects and is used to create a wide range of structures, including buildings, walls, and other infrastructure projects.



Figure 5 : Brick Masonry Work

<https://theconstructor.org/building/brick-masonry-definition-types-construction/25916/>

9. PLASTERING WORK:

Plastering is a construction technique that involves the application of a thin layer of plaster or other material to a surface in order to create a smooth and even finish. Plastering is commonly used on walls, ceilings, and other surfaces to create a smooth, finished appearance and to hide any imperfections or irregularities. The process of plastering involves applying a wet mixture of plaster, water, and other materials to the surface using a trowel or other tool. The plaster is then smoothed and finished to create a smooth, even surface. Plastering is an important aspect of many construction projects and is used to create a smooth, finished appearance on walls, ceilings, and other surfaces. It requires skill and attention to detail to ensure that the work is completed to the required standards.



Figure 6 : Plastering Work

<http://gulfshine.ae/service/plaster-works/>

10. WOOD SHUTTER:

The wood sections used in frames and shutters should be 16 microns thick and coated with powder in an approved color. All the frames and shutters should be the same color. The contractor should verify the exact dimensions at the site before fabrication. The measurement should be in square meters of the window area actually installed on site, including fixed,



sliding, and openable shutters.

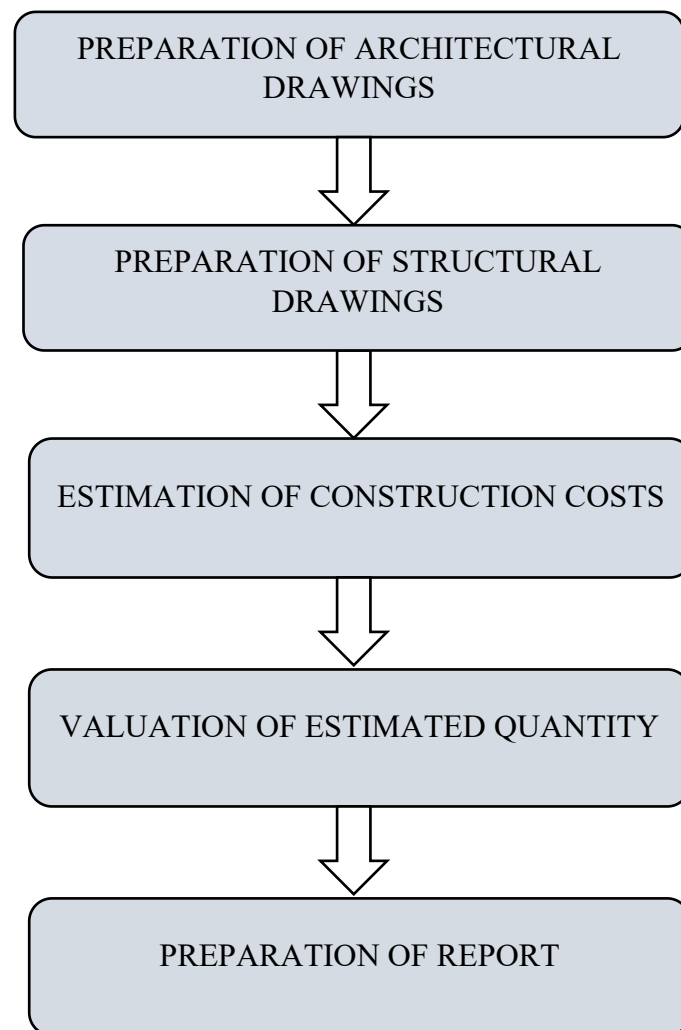
Figure 7 : Wood Shutter

<http://muskaninterior.com.np/Wood-window>

C. METHODOLOGY

The project consists of three main steps:

1. Drawing the architectural and structural components of the proposed building.
2. Calculating the detailed quantity of major components of the building.
3. Finding out the cost of construction by assigning the estimated quantity with viable rates according to the prevailing rates for the given area.



D. DRAWINGS

1. BUILDING LOCATION

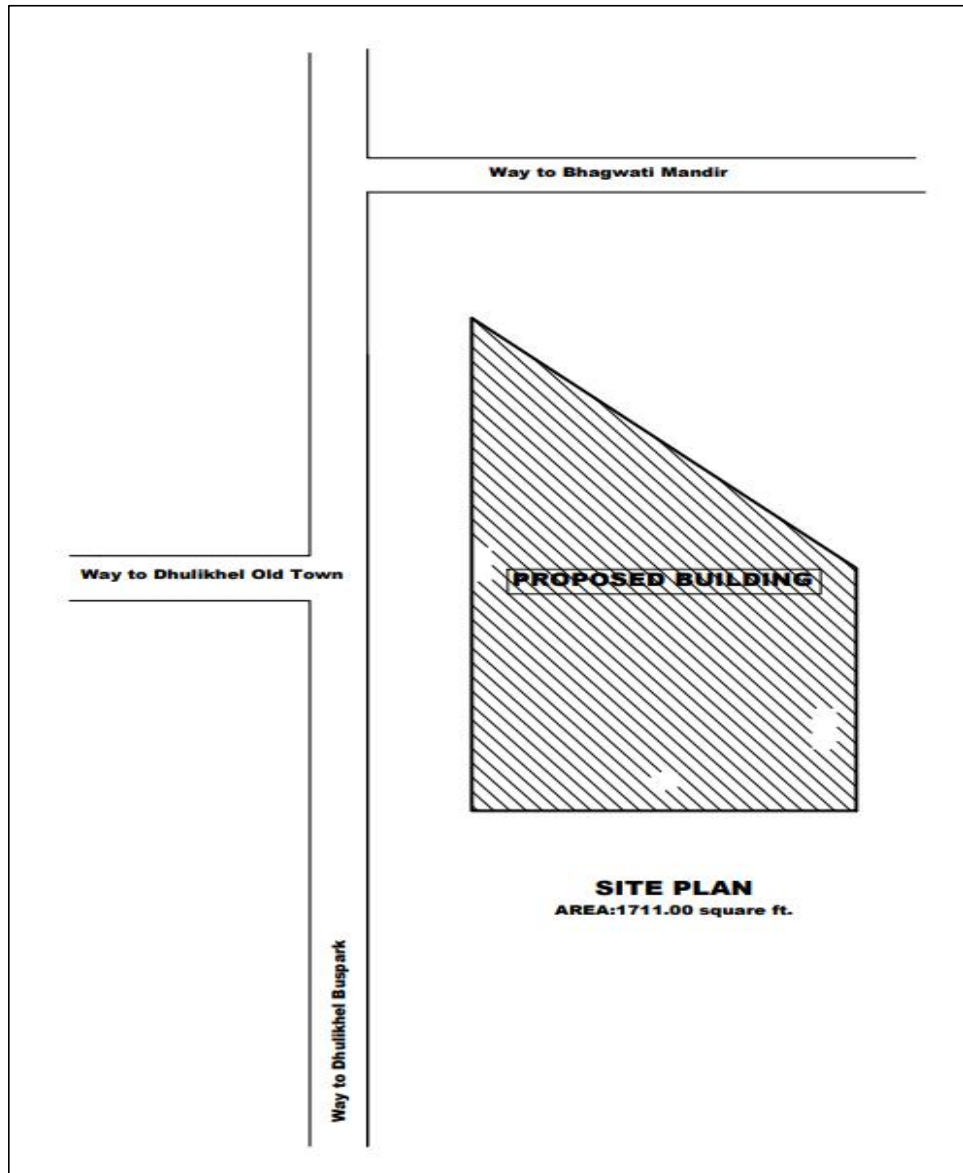


Figure 8 : Building Site Map

2. ARCHITECTURAL DRAWINGS:

The architectural drawing consists of floor plans, elevations and sectional part of any buildings. These drawings are done as per the architectural design codes in the national building code.

FLOOR PLANS:

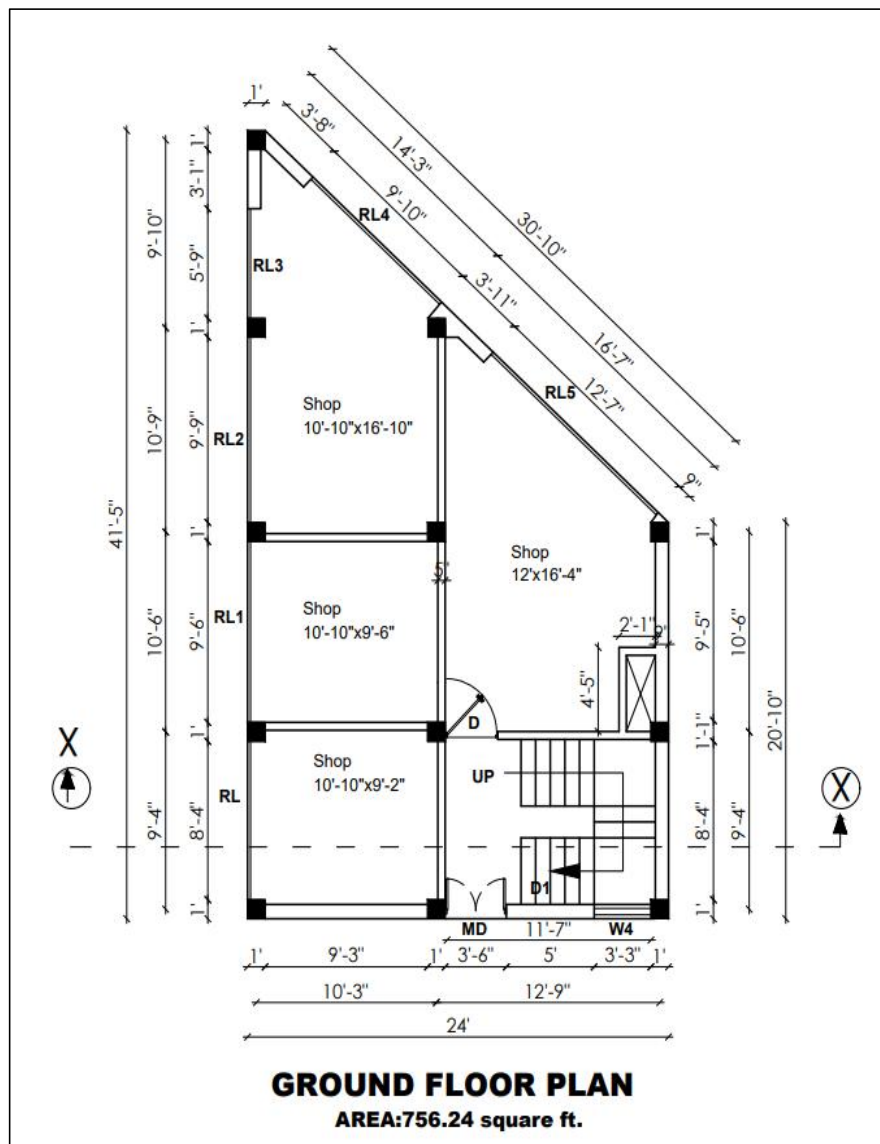


Figure 9 : Ground Floor Plan

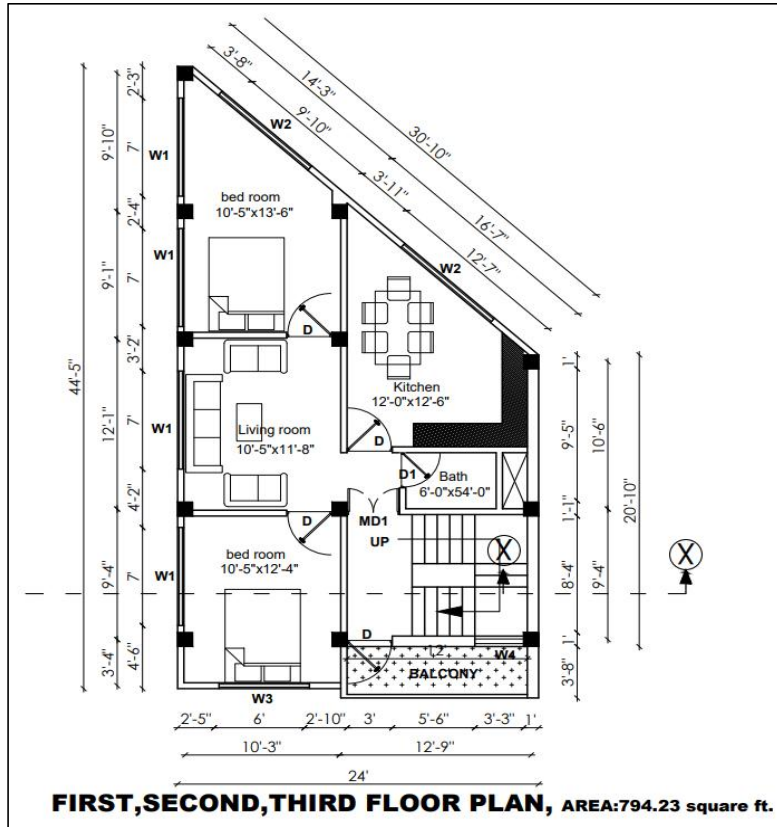


Figure 10 : 1st, 2nd & 3rd Floor Plan

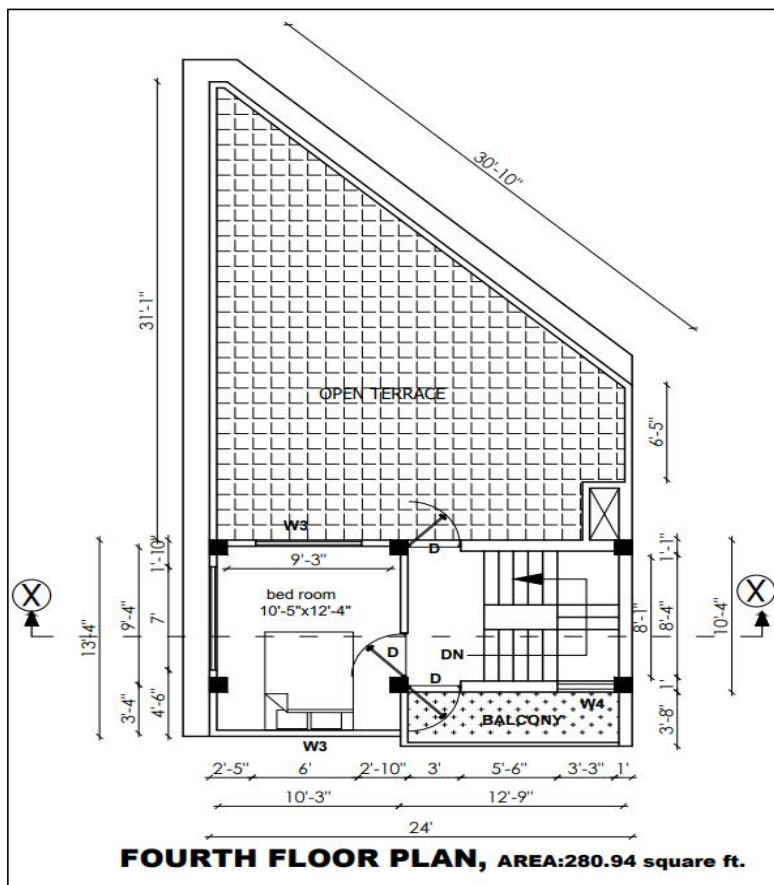


Figure 11 : 4th Floor Plan

SECTIONAL ELEVATION:

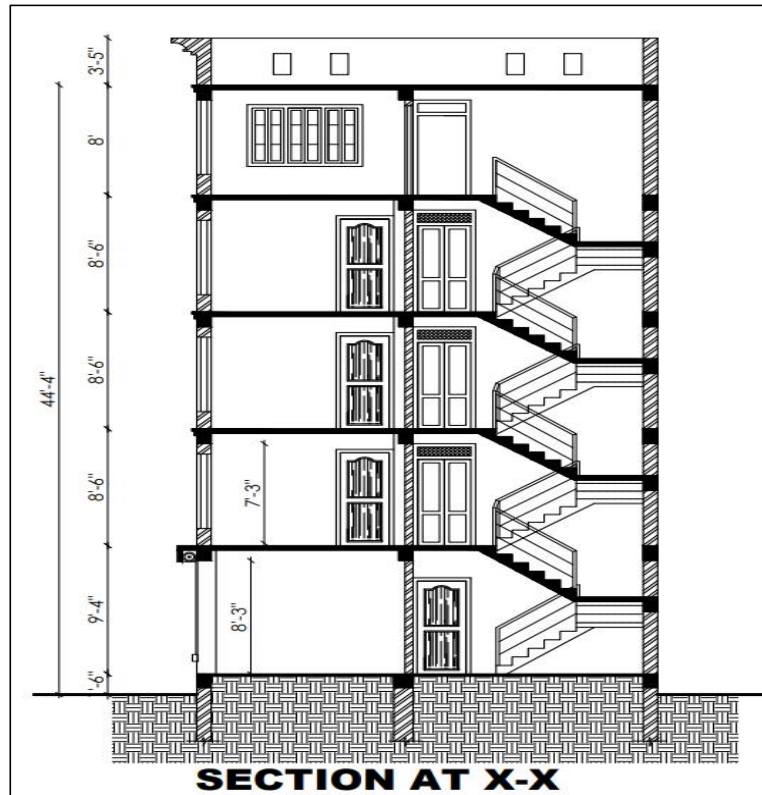


Figure 12 : Sectional Elevation

Table 3 : Opening Schedule

OPENING SCHEDULE				
SN	PARTICULAR	SYMBOL	SIZE(LXH)	NOS
1.	WINDOW	W1	7'-0"X6'-0"	13
2.	WINDOW	W2	8'-0"X6'-0"	6
3.	WINDOW	W3	6'-0"X6'-0"	5
4	WINDOW	W4	3'-3"X4'-0"	5
5	MAIN DOOR	MD	3'-6"X8'-3"	1
7	MAIN DOOR	MD1	3'-3"X8'-3"	3
8	DOOR	D	3'-0"X7'-6"	15
9	DOOR	D1	2'-6"X6'-6"	4
10	ROLLING SHUTTER	RL	8'-4"X8'-3"	1
11	ROLLING SHUTTER	RL1	9'-6"X8'-3"	1
12	ROLLING SHUTTER	RL2	9'-9"X8'-3"	1
13	ROLLING SHUTTER	RL3	5'-9"X8'-3"	1
14	ROLLING SHUTTER	RL4	9'-10"X8'-3"	1
15	ROLLING SHUTTER	RL5	12'-7"X8'-3"	1

ELEVATIONS:

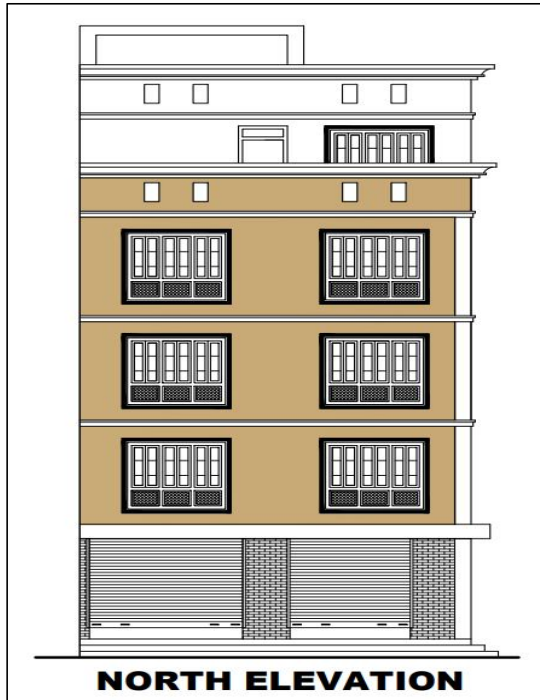


Figure 13 : Elevations

3. STRUCTURAL DRAWINGS:

The structural drawings consist of detailed layout and reinforcement details of all the structural components of the building including footings, columns, beams, slab, staircase, lintels. etc. Below shown are the structural drawings based on which the estimation of the entire building is carried out.

TRENCH PLAN:

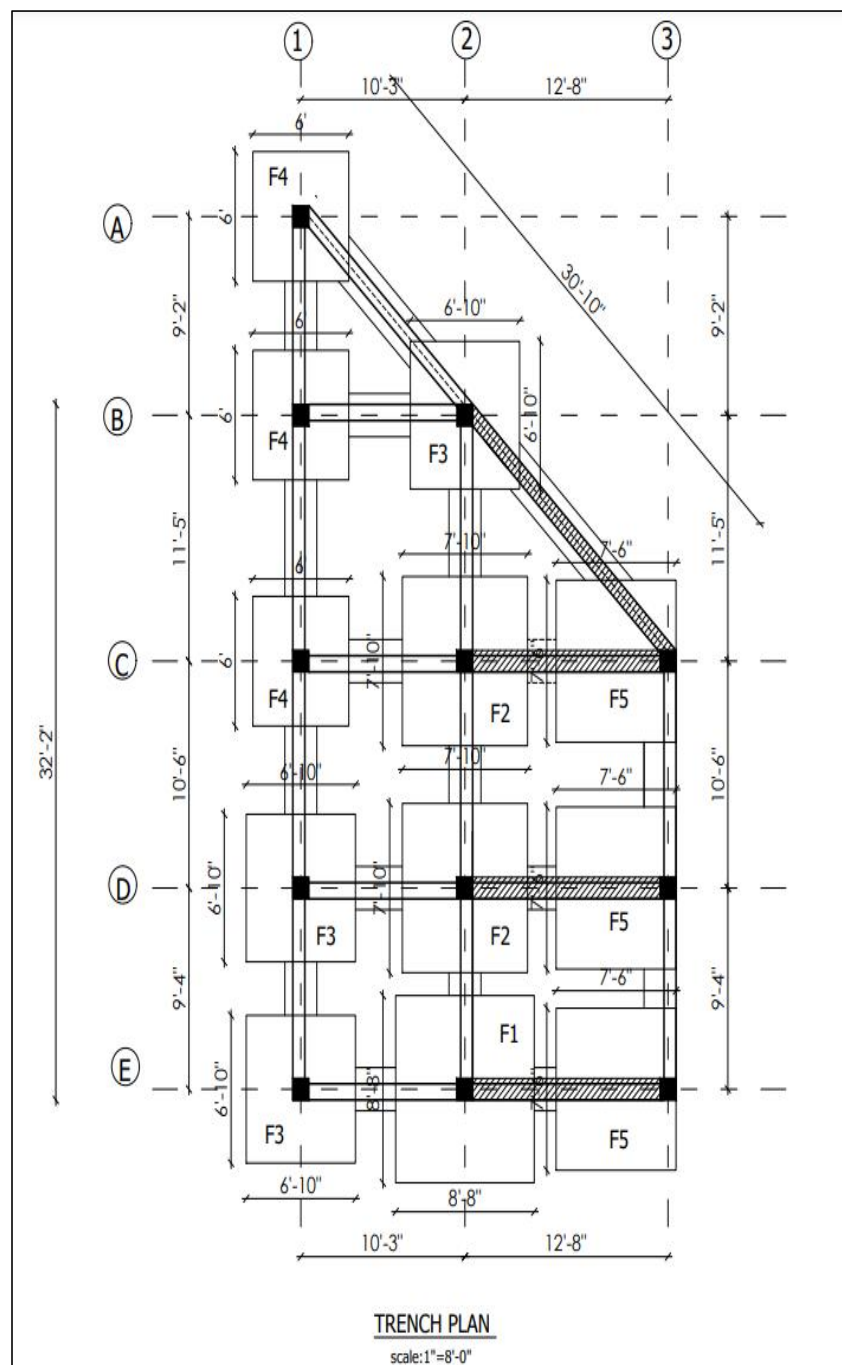


Figure 14 : Trench Plan

COLUMN LAYOUT PLAN:

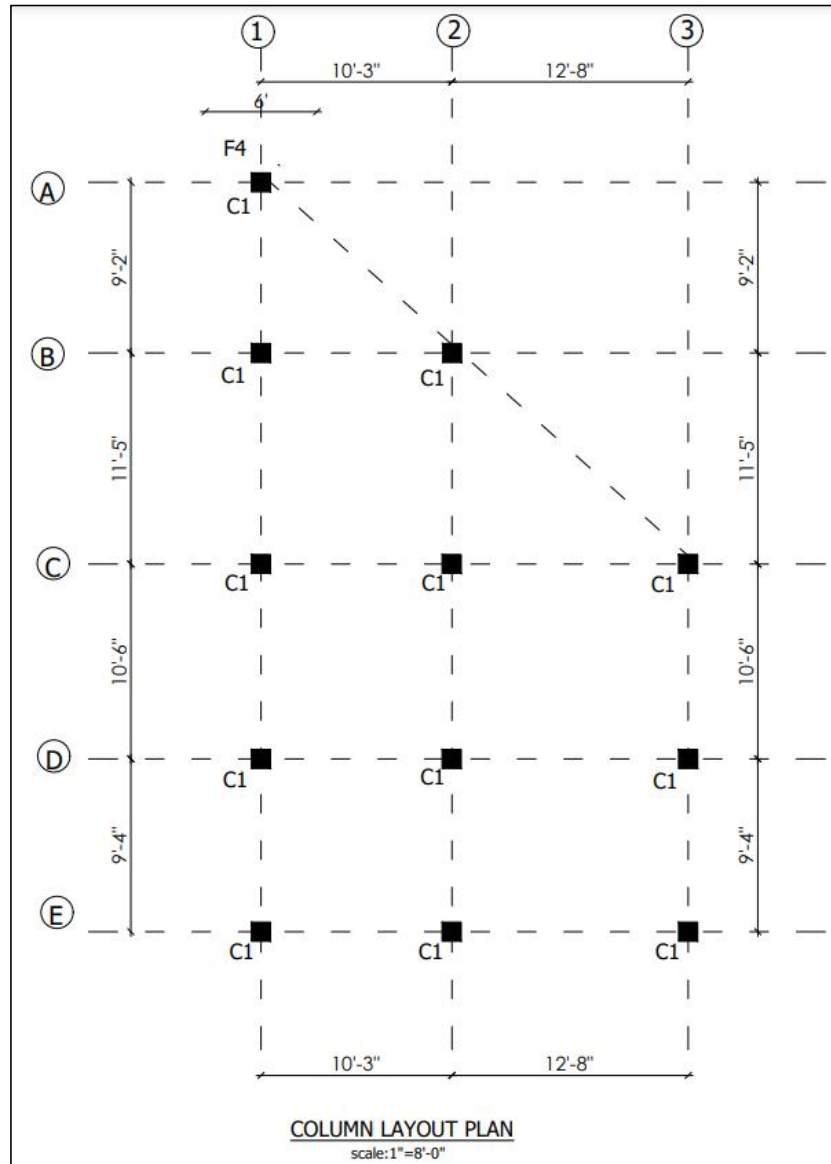


Figure 15 : Column Layout Plan

FOUNDATION PLAN AND SECTION:

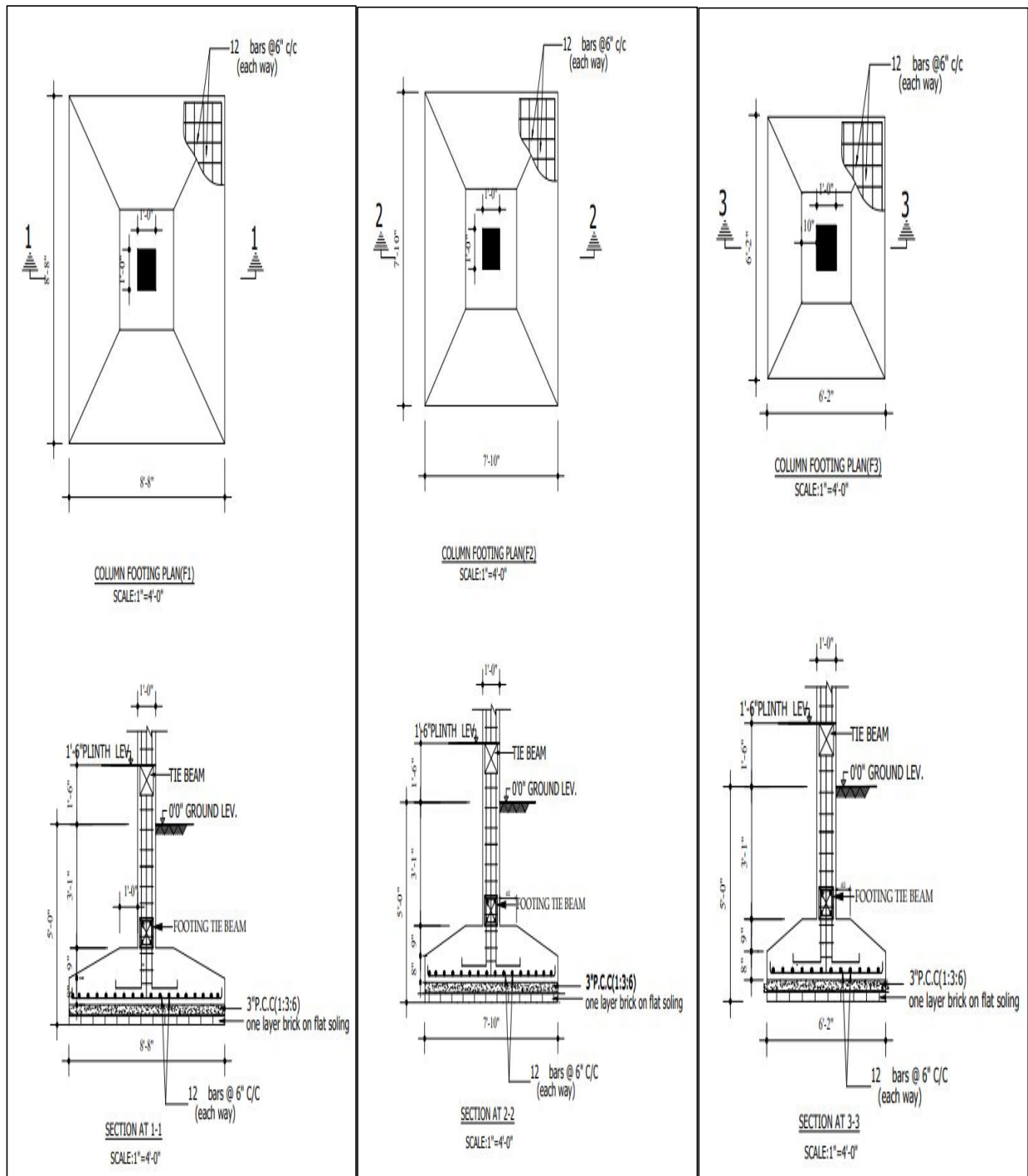


Figure 16 : Foundation Plans and Sections

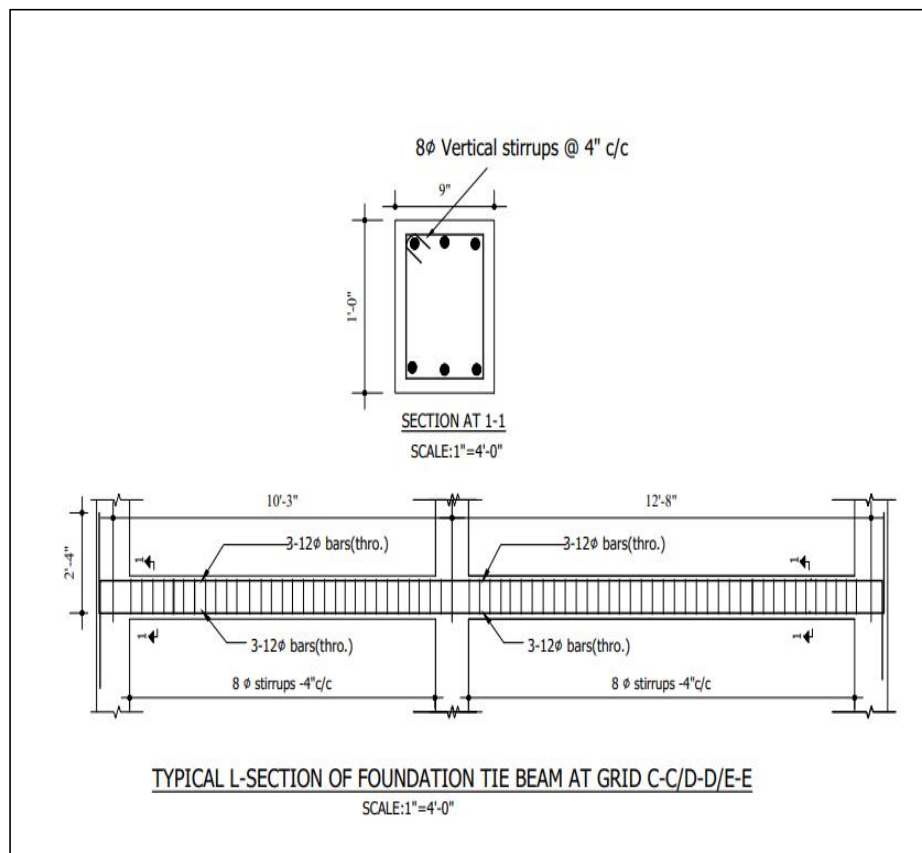
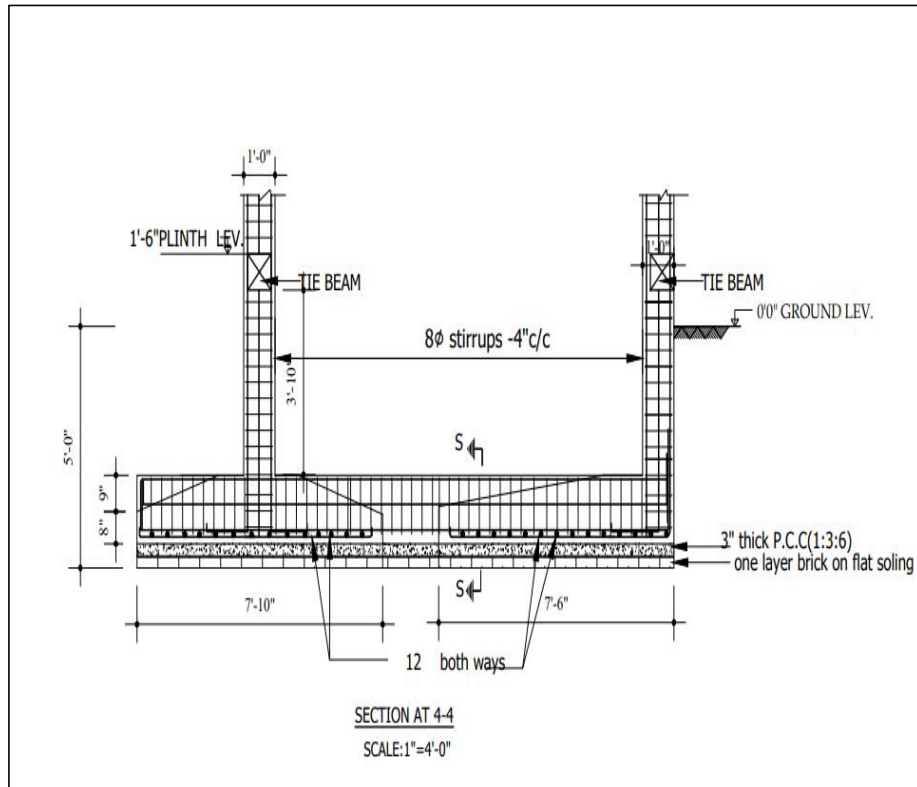


Figure 17 : Sections of Foundation Beam

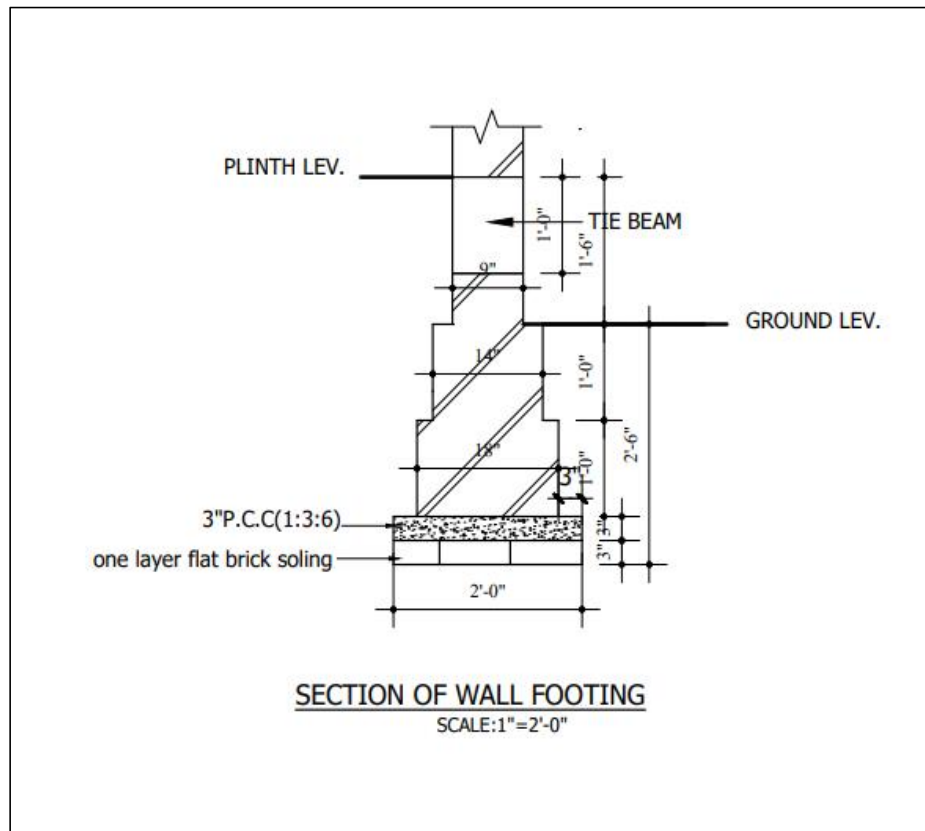


Figure 18 : Section of Wall Footing

COLUMN DETAILS

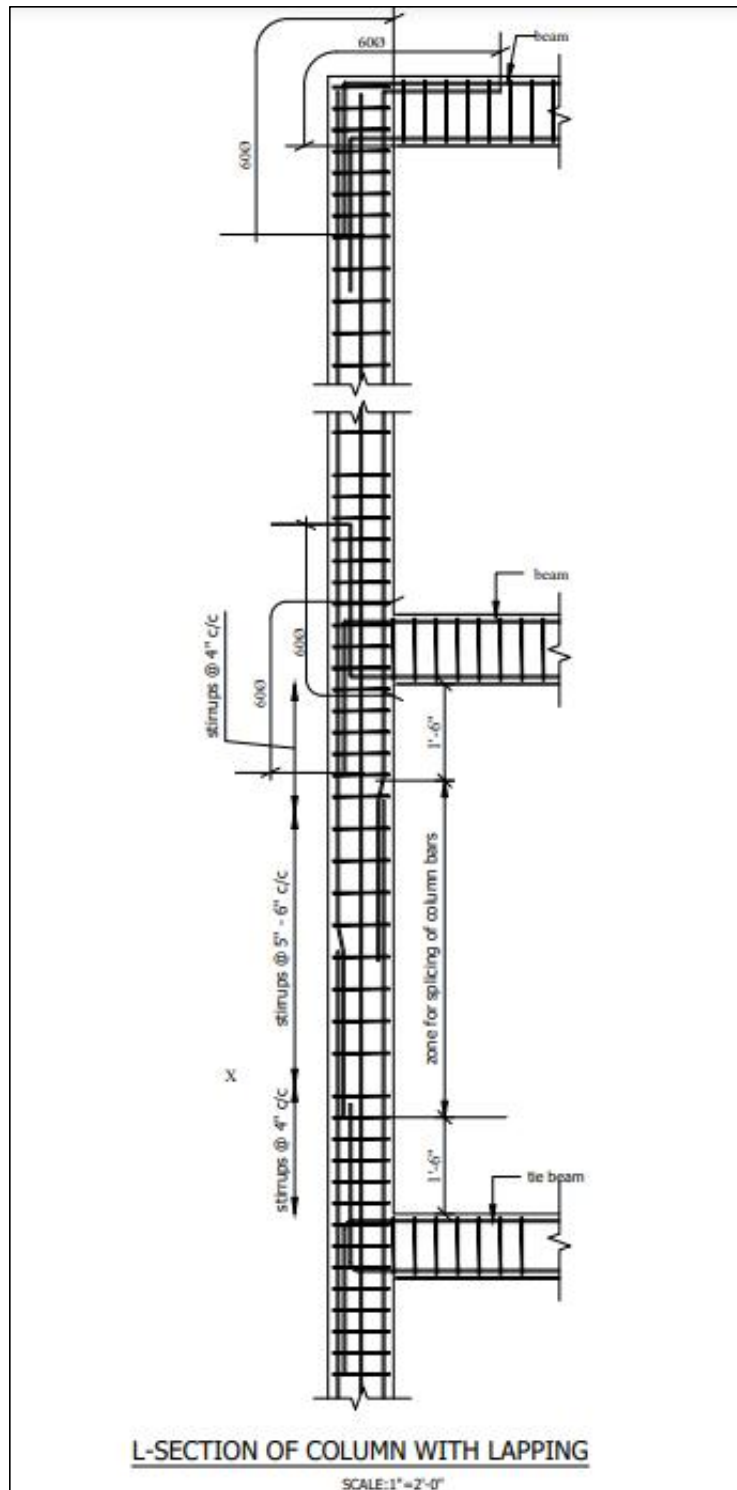
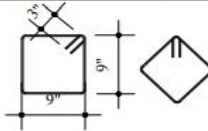


Figure 19 : Column Lapping with Beams

COLUMN SCHEDULE

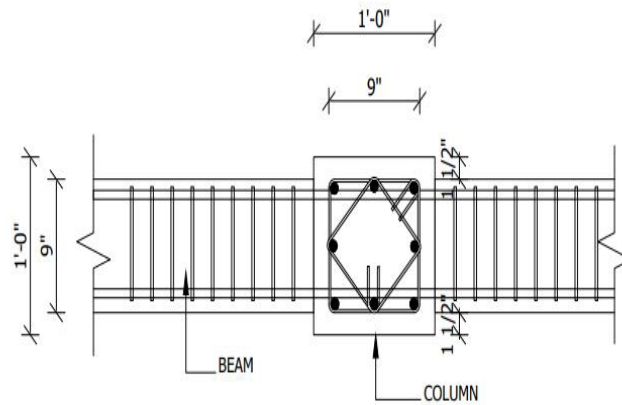
COLUMN TYPE	C1(12"x12")
FLOOR	REINFORCEMENT
SEMI-BASEMENT FLOOR	 8-16
GROUND FLOOR	 8-16
FIRST FLOOR	 4-16 +4-12
SECOND FLOOR	 4-16 +4-12

STIRRUPS DETAILS



stirrups, 8 4"/4" at end ties.
stirrups, 8 4"/4" at mid ties.

CLOSED TIES FOR 12"x12" COLUMN



BEAM & COLUMN JUNCTION DETAIL PLAN

12"x12" COLUMN
SCALE: 1"=1'-0"

Figure 20 : Beam and Column Junction

SLAB REINFORCEMENT:

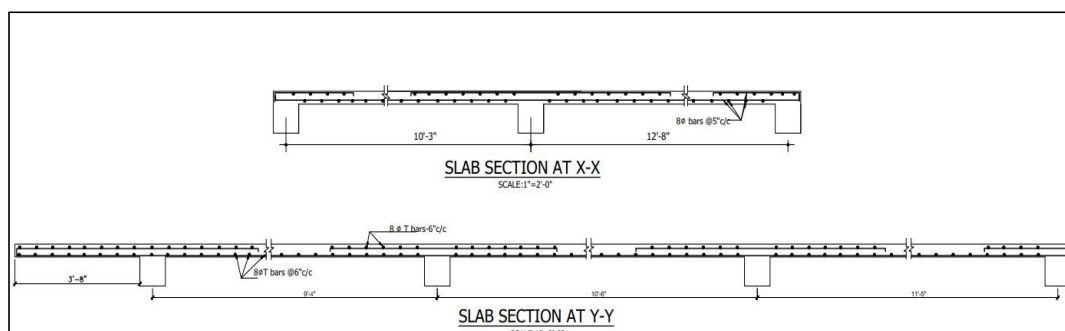
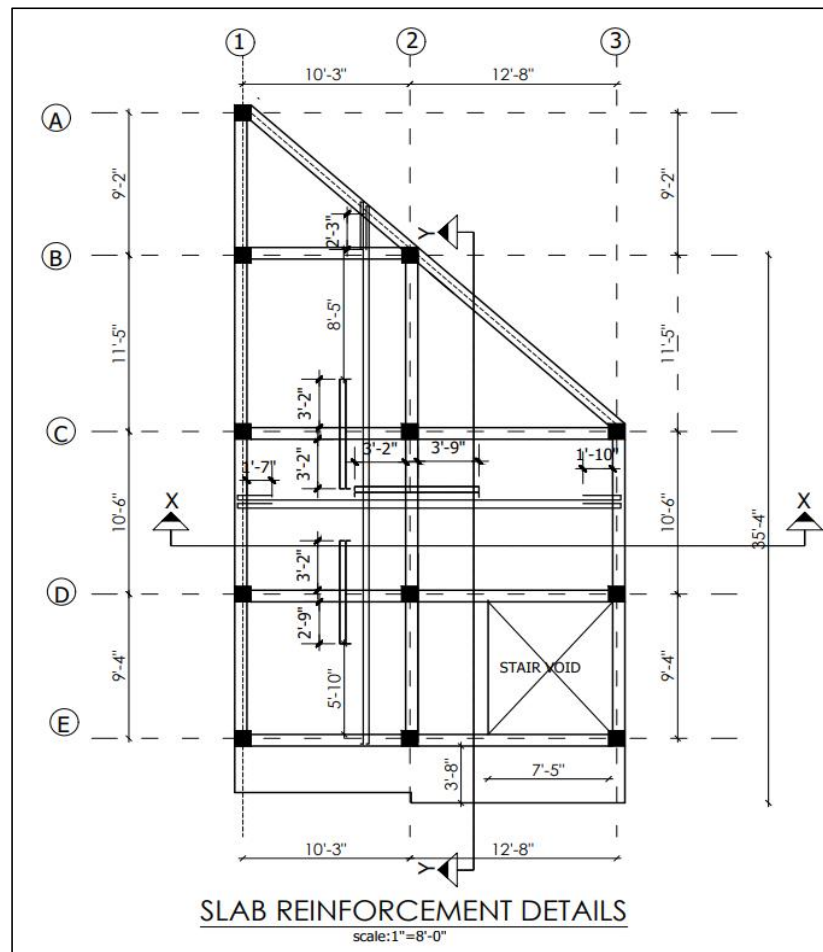


Figure 21 : Slab Reinforcement Details

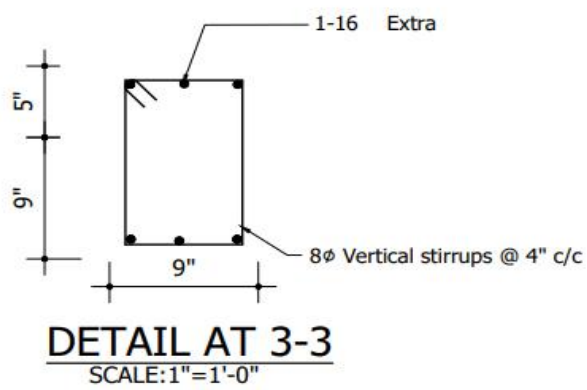
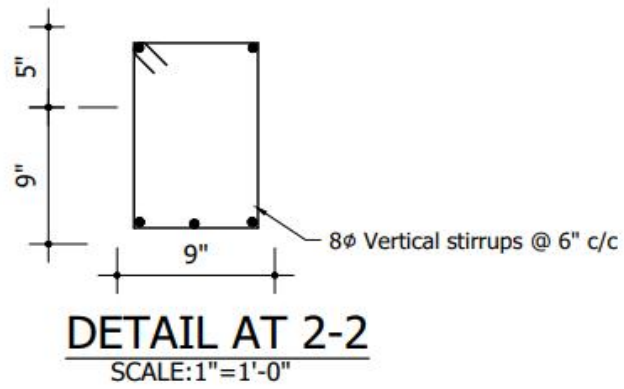
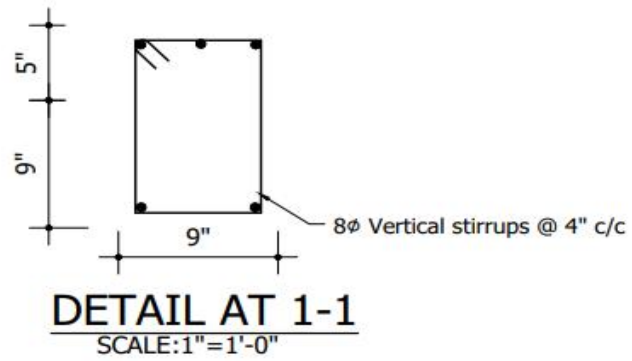


Figure 22 : Stirrups Details

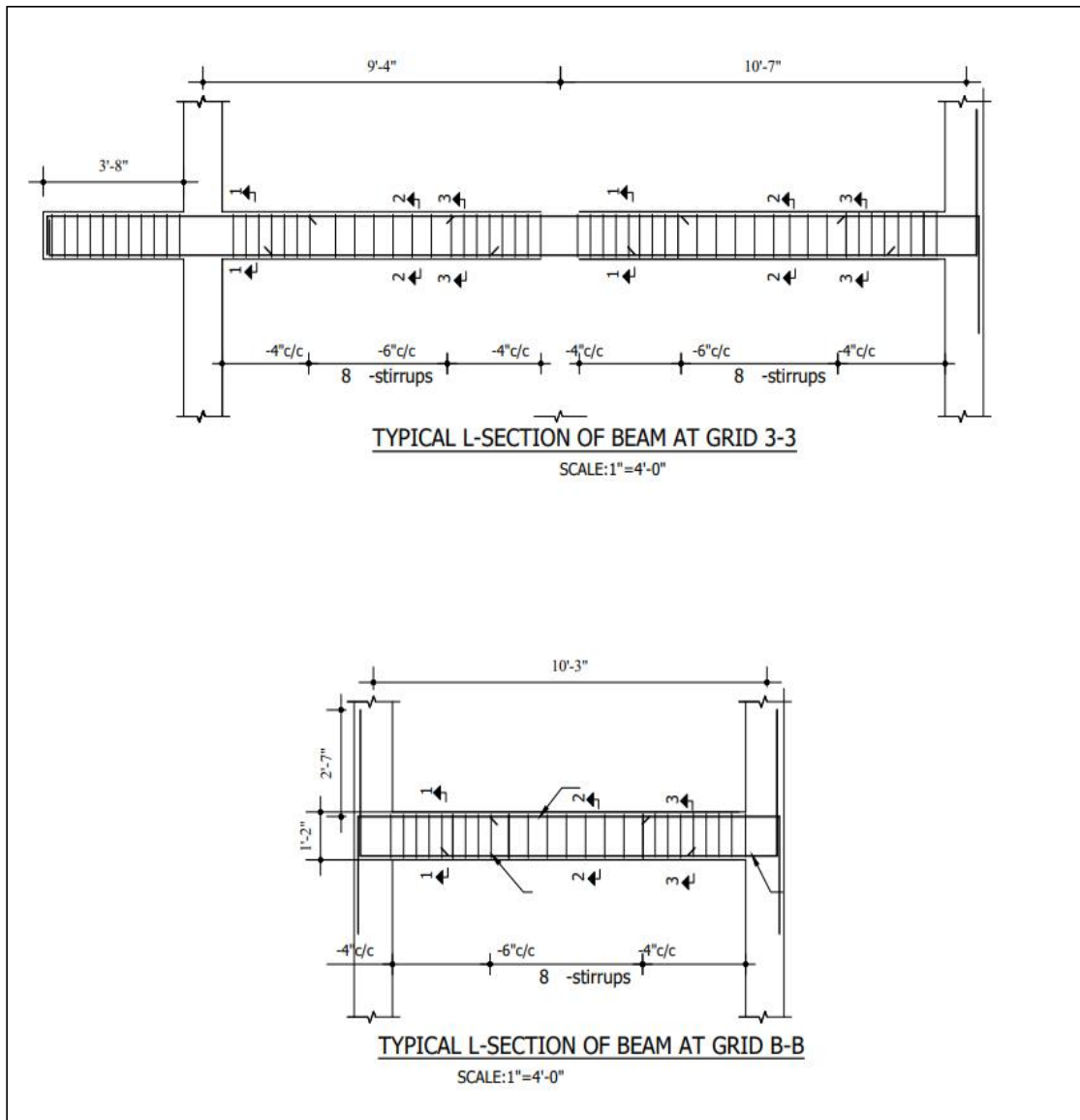


Figure 23 : L Section of Slab Beam at different sections

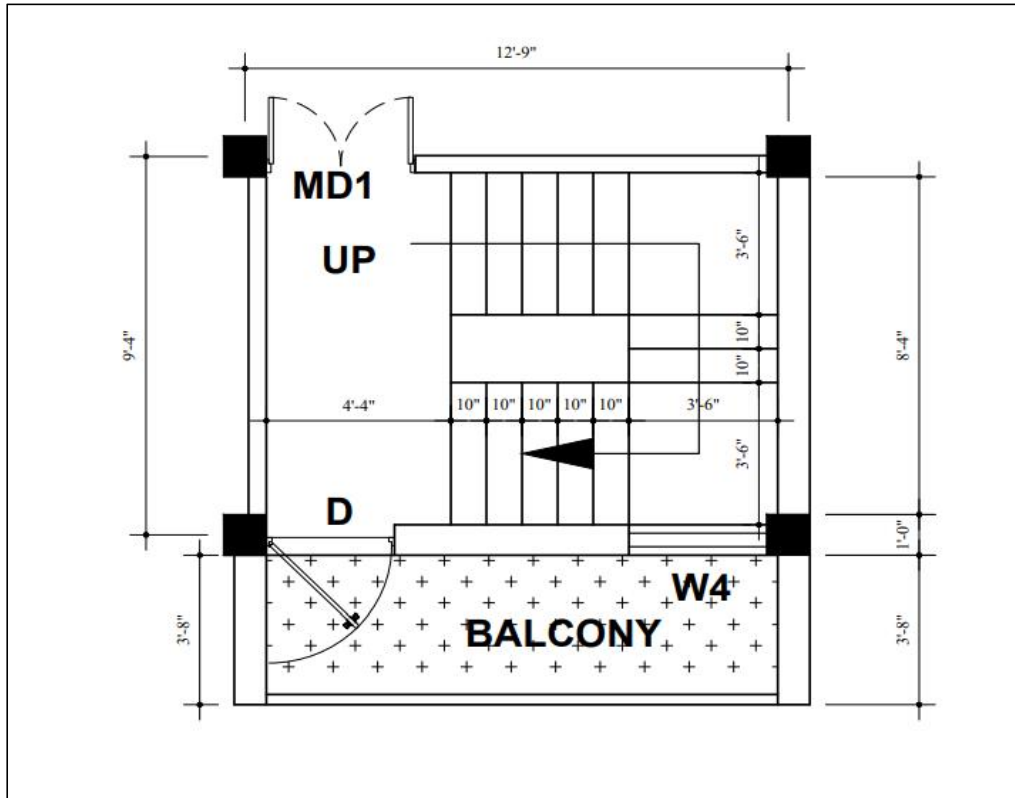


Figure 24 : Staircase Plan

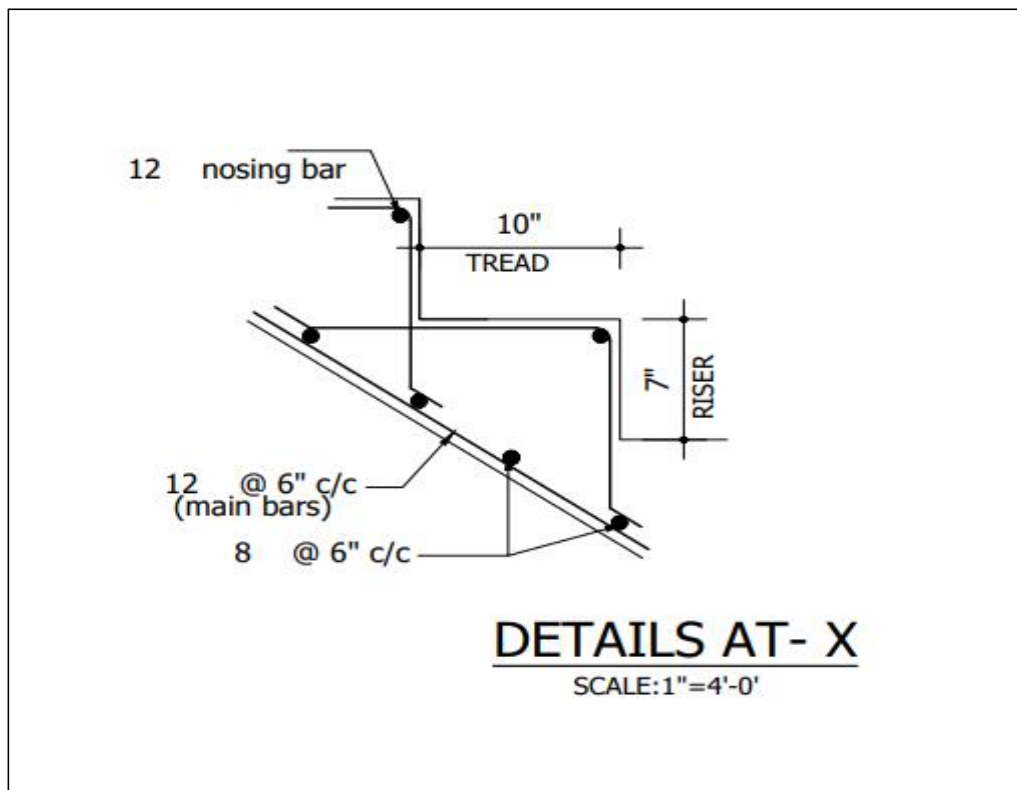


Figure 25 : Riser and Treader Details

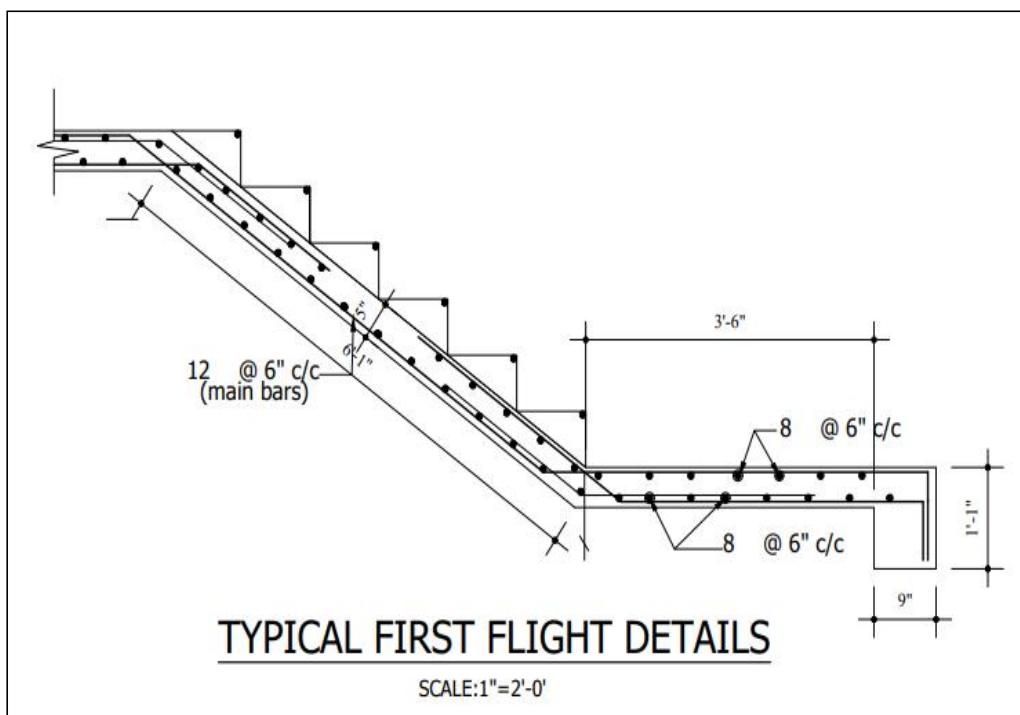
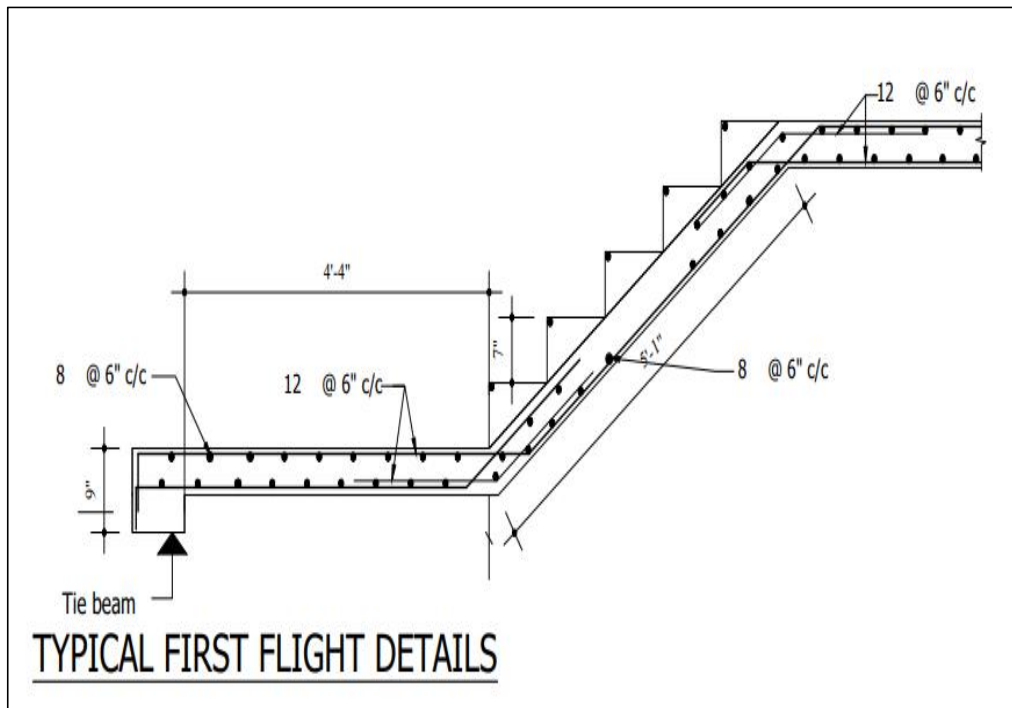


Figure 26 : Section of First Flight

WALL:

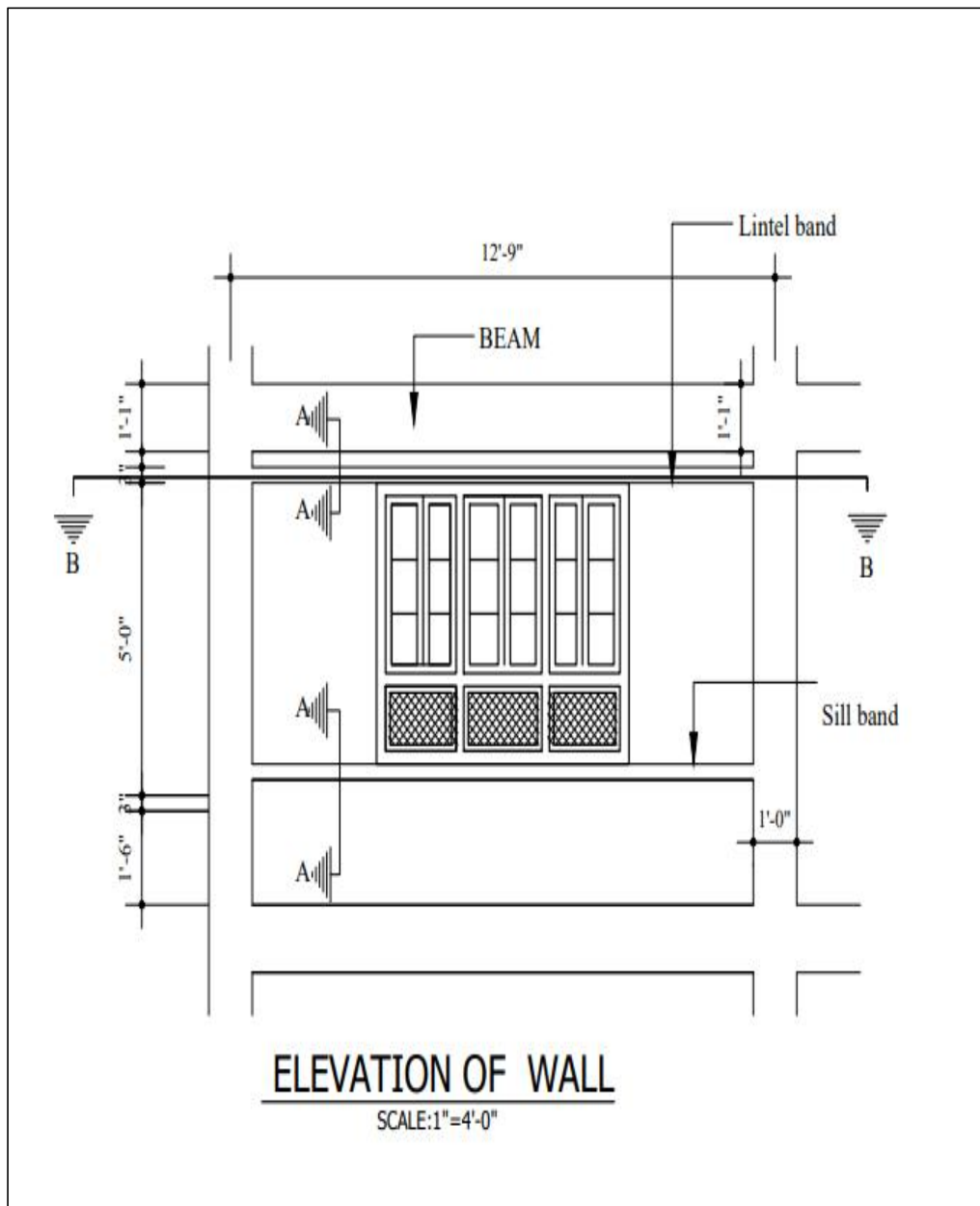


Figure 27 : Elevation of Wall

E. CALCULATION TABLE:

Description of work	No	Length	Breadth	Height	Quantity
Excavation of work					
Footing					
F1	1	8.66	8.66	5.00	374.98
F2	2	7.83	7.83	5.00	613.09
F3	3	6.83	6.83	5.00	699.73
F4	3	6.00	6.00	5.00	540.00
F5	3	7.50	7.50	5.00	843.75
		Sub total			3071.55

Table 4 : Earthwork Excavation

Description of work	No	Length	Breadth	Height	Quantity	Unit
Soling work						
F1	1	8.66	8.66		75.00	sq ft
F2	2	7.83	7.83		122.62	sq ft
F3	3	6.83	6.83		139.95	sq ft
F4	3	6.00	6.00		108.00	sq ft
F5	3	7.50	7.50		168.75	sq ft
Soling work Floor	1	745.00	1.00		745.00	sq ft
	Total				1359.31	sq ft

Table 5 : Soiling work

Description of work	No	Length	Breadth	Height	Quantity	Unit
PCC work						
F1	1	8.66	8.66	0.25	18.75	cu. ft.
F2	2	7.83	7.83	0.25	30.65	cu. ft.
F3	3	6.83	6.83	0.25	34.99	cu. ft.
F4	3	6.00	6.00	0.25	27.00	cu. ft.
F5	3	7.50	7.50	0.25	42.19	cu. ft.
Soling work Floor	1	745.00	1.00	0.25	186.25	cu. ft.
	Total				339.83	cu. ft.

Description of work	No	Length	Breadth	Height	Quantity
1)Footing	Area				
<i>F1</i>	755.00			4.50	3,397.50
		Sub total			3,397.50

Table 6 : PCC and Soil filling Work

RCC WORK:

Description of work	No	Length	Breadth	Height	Quantity	Unit
Rectangle portion						
F1	1	8.66	8.66	0.66	49.50	cu. ft.
F2	2	7.83	7.83	0.66	80.93	cu. ft.
F3	3	6.83	6.83	0.66	92.36	cu. ft.
F4	3	6.00	6.00	0.66	71.28	cu. ft.
F5	3	7.50	7.50	0.66	111.38	cu. ft.
Trapezoidal portion						
F1	1	5.58	5.58	0.75	23.35	cu. ft.
F2	2	5.16	5.16	0.75	39.94	cu. ft.
F3	3	4.66	4.66	0.75	48.86	cu. ft.
F4	3	4.25	4.25	0.75	40.64	cu. ft.
F5	3	5.00	5.00	0.75	56.25	cu. ft.
Plinth level Beam						
,C-C,D-D,E-E	4	20.91	0.75	1.00	62.73	cu. ft.
A-A	1	29.83	0.75	1.00	22.37	cu. ft.
B-B	1	9.25	0.75	1.00	6.94	cu. ft.
GriD1-1	1	36.41	0.75	2.00	54.62	cu. ft.
Grid2-2	1	28.00	0.75	3.00	63.00	cu. ft.
Grid3-3	1	17.83	0.75	4.00	53.49	cu. ft.
Pedestal Column	12	1.00	1.00	3.83	45.96	cu. ft.
Ground floor						
Column	12	1.00	1.00	9.00	108.00	cu. ft.
Beam					0.00	cu. ft.
,C-C,D-D,E-E	4	20.91	0.75	0.75	47.05	cu. ft.
A-A	1	29.83	0.75	0.75	16.78	cu. ft.
B-B	1	9.25	0.75	0.75	5.20	cu. ft.
GriD1-1	1	36.41	0.75	0.75	20.48	cu. ft.
Grid2-2	1	28.00	0.75	0.75	15.75	cu. ft.
Grid3-3	1	17.83	0.75	0.75	10.03	cu. ft.
Slab	1	1.00	0.41	745.51	305.66	cu. ft.
Staircase					0.00	cu. ft.
First Floor & Third Flight	2	5.50	3.50	0.41	15.79	cu. ft.
Second flight	1	2.50	3.50	0.41	3.59	cu. ft.
First & Third Step	10	3.50	1.00	0.24	8.40	cu. ft.
Second Step	2	3.50	1.00	0.24	1.68	cu. ft.
Landing					0.00	cu. ft.
First & Second	2	3.50	3.50	0.41	10.05	cu. ft.
Deduction part	-1	8.33	8.33	0.41	-28.45	cu. ft.

First floor						
Column	12	1.00	1.00	8.08	96.96	cu. ft.
Beam						
,C-C,D-D,E-E	4	20.91	0.75	0.75	47.05	cu. ft.
A-A	1	29.83	0.75	0.75	16.78	cu. ft.
B-B	1	9.25	0.75	0.75	5.20	cu. ft.
GriD1-1	1	36.41	0.75	0.75	20.48	cu. ft.
Grid2-2	1	28.00	0.75	0.75	15.75	cu. ft.
Grid3-3	1	17.83	0.75	0.75	10.03	cu. ft.
Slab	1	1.00	0.41	826.00	338.66	cu. ft.
Staircase						
First floor &Third flight	2	5.50	3.50	0.41	15.79	cu. ft.
Second Flight	1	2.50	3.50	0.41	3.59	cu. ft.
First & Third Step	10	3.50	1.00	0.24	8.40	cu. ft.
Second Step	2	3.50	1.00	0.24	1.68	cu. ft.
Landing					0.00	cu. ft.
First & Second	2	3.50	3.50	0.41	10.05	cu. ft.
Deduction part	-1	8.33	8.33	0.41	-28.45	cu. ft.
Second floor						
Column	12	1.00	1.00	8.09	97.08	cu. ft.
Beam						
,C-C,D-D,E-E	4	20.91	0.75	0.75	47.05	cu. ft.
A-A	1	29.83	0.75	0.75	16.78	cu. ft.
B-B	1	9.25	0.75	0.75	5.20	cu. ft.
GriD1-1	1	36.41	0.75	0.75	20.48	cu. ft.
Grid2-2	1	28.00	0.75	0.75	15.75	cu. ft.
Grid3-3	1	17.83	0.75	0.75	10.03	cu. ft.
Slab	1	1.00	0.41	826.00	338.66	cu. ft.

Staircase						
First floor &Third flight	2	5.50	3.50	0.41	15.79	cu. ft.
Second flight	1	2.50	3.50	0.41	3.59	cu. ft.
First & Third Step	10	3.50	1.00	0.24	8.40	cu. ft.
Second Step	2	3.50	1.00	0.24	1.68	cu. ft.
Landing					0.00	cu. ft.
First & Second	2	3.50	3.50	0.41	10.05	cu. ft.
Deduction part	-1	8.33	8.33	0.41	-28.45	cu. ft.

Third floor						
Column	12	1.00	1.00	8.00	96.00	cu. ft.
Beam						
,C-C,D-D,E-E	4	20.91	0.75	0.75	47.05	cu. ft.
A-A	1	29.83	0.75	0.75	16.78	cu. ft.
B-B	1	9.25	0.75	0.75	5.20	cu. ft.
Grid1-1	1	36.41	0.75	0.75	20.48	cu. ft.
Grid2-2	1	28.00	0.75	0.75	15.75	cu. ft.
Grid3-3	1	17.83	0.75	0.75	10.03	cu. ft.
Slab	1	1.00	0.41	826.00	338.66	cu. ft.
Staircase						
First floor &Third flight	2	5.50	3.50	0.41	15.79	cu. ft.
Second flight	1	2.50	3.50	0.41	3.59	cu. ft.
First & Third Step	10	3.50	1.00	0.24	8.40	cu. ft.
Second Step	2	3.50	1.00	0.24	1.68	cu. ft.
Landing					0.00	cu. ft.
First & Second	2	3.50	3.50	0.41	10.05	cu. ft.
Deduction part	-1	8.33	8.33	0.41	-28.45	cu. ft.

Fourth floor						
Slab	1	1.00	327.00	0.41	134.07	cu. ft.
Top	1	45.00	1.50	0.41	27.68	cu. ft.
Top	1	30.00	1.50	0.41	18.45	cu. ft.
Beam	2	24.00	0.75	0.75	27.00	cu. ft.
	3	13.00	0.75	0.75	21.94	cu. ft.
Column	6	1.00	1.00	8.09	48.54	cu. ft.
		Sub total			3985.00	cu. ft.

Table 7 : RCC Work

REINFORCEMENT:

Description of work	No	Dia (mm)	Structure No	Length F	Lenth in M	Weight	Quantity	Unit
Footing								
<i>F1</i>								
X-axis	1	12.00	18.00	8.75	2.66	0.89	42.56	kg.
Y -axis	1	12.00	18.00	8.75	2.66	0.89	42.56	kg.
<i>F2</i>						0.00		
X-axis	2	12.00	16.00	8.16	2.48	0.89	70.56	kg.
Y -axis	2	12.00	16.00	8.16	2.48	0.89	70.56	kg.
<i>F3</i>								
X-axis	3	12.00	14.00	7.16	2.18	0.89	81.26	kg.
Y -axis	3	12.00	14.00	8.16	2.48	0.89	92.61	kg.
<i>F4</i>								
X-axis	3	12.00	12.00	7.33	2.23	0.89	71.31	kg.
Y -axis	3	12.00	12.00	7.33	2.23	0.89	71.31	kg.
<i>F5</i>								
X-axis	3	12.00	16.00	7.83	2.38	0.89	101.56	kg.
Y -axis	3	12.00	16.00	7.83	2.38	0.89	101.56	kg.
Plinth Level Beam			1.00					
Grid C-C,D-D,E-E								
Top	3	12.00	3.00	25.91	7.88	0.89	63.01	kg.
Bottom	3	12.00	3.00	27.91	8.48	0.89	67.88	kg.
<i>Grid B-B</i>								
Top	1	12.00	3.00	12.25	3.72	0.89	9.93	kg.
Bottom	1	12.00	3.00	14.50	4.41	0.89	11.75	kg.
<i>Grid A- A</i>								
Top	1	12.00	3.00	32.75	9.96	0.89	26.55	kg.
Bottom	1	12.00	3.00	36.75	11.17	0.89	29.79	kg.
Grid 1-1								
Top	1	12.00	3.00	43.50	13.22	0.89	35.26	kg.
Bottom	1	12.00	3.00	45.50	13.83	0.89	36.89	kg.
Grid 2-2								kg.
Top	1	12.00	3.00	34.16	10.38	0.89	27.69	kg.

Bottom	1	12.00	3.00	36.16	10.99	0.89	29.31	kg.
Grid 3-3								
Top	1	12.00	3.00	22.83	6.94	0.89	18.51	kg.
Bottom	1	12.00	3.00	24.83	7.55	0.89	20.13	kg.
Column	12	16.00	8.00	19.13	5.82	1.58	882.24	kg.
Plinth level Beam								
Grid1-1	1	8.00	124.00	3.16	0.96	0.40	47.06	kg.
Grid2-2	1	8.00	96.00	3.16	0.96	0.40	36.43	kg.
Grid3-3	1	8.00	63.00	3.16	0.96	0.40	23.91	kg.
Grid C-C,D-D,E-E	3	8.00	72.00	3.16	0.96	0.40	81.97	kg.
A-A	1	8.00	93.00	3.16	0.96	0.40	35.29	kg.
B-B	1	8.00	30.00	3.16	0.96	0.40	11.39	kg.
				Up to Tie Beam				

Beam								
Ground floor								
Grid C-C,D-D,E-E								
	3	16.00	4.00	25.91	7.88	1.58	149.36	kg.
A-A,3 triangle	1	16.00	4.00	36.00	10.94	1.58	69.18	kg.
B-B	1	16.00	4.00	14.00	4.26	1.58	26.90	kg.
Grid 1-1	1	16.00	4.00	45.00	13.68	1.58	86.47	kg.
Grid 2-2	1	16.00	4.00	36.00	10.94	1.58	69.18	kg.
Grid 3-3	1	16.00	4.00	25.00	7.60	1.58	48.04	kg.
Slab								
Down layer								
A-3	1	8.00	72.00	24.00	7.30	0.40	207.53	kg.
A-E	1	8.00	50.00	36.00	10.94	0.40	216.18	kg.
Top layer								
A-3	1	8.00	36.00	23.00	6.99	0.40	99.44	kg.
A-E	1	8.00	21.00	35.00	10.64	0.40	88.27	kg.
Extra Bar	1	8.00	35.00	2.00	0.61	0.40	8.41	kg.
	1	8.00	35.00	2.00	0.61	0.40	8.41	kg.
	1	8.00	35.00	7.00	2.13	0.40	29.42	kg.
	1	8.00	10.00	7.16	2.18	0.40	8.60	kg.
	1	8.00	14.00	4.41	1.34	0.40	7.41	kg.
	1	8.00	10.00	2.25	0.68	0.40	2.70	kg.
	1	8.00	14.00	2.25	0.68	0.40	3.78	kg.
Deduction	1	8.00	10.00	2.25	0.68	0.40	2.70	kg.

	-2	8.00	18.00	7.41	2.25	0.40	-32.04	kg.
	-2	8.00	14.00	9.33	2.84	0.40	-31.37	kg.
Stair case								
First flight & second								
Main Bar	2	12.00	7.00	17.00	5.17	0.89	64.31	kg.
	2	12.00	7.00	7.00	2.13	0.89	26.48	kg.
	2	12.00	7.00	5.00	1.52	0.89	18.92	kg.
	2	12.00	7.00	3.00	0.91	0.89	11.35	kg.
	2	12.00	7.00	3.50	1.06	0.89	13.24	kg.
Distribution bar	2	8.00	34.00	3.50	1.06	0.40	28.58	kg.
	2	8.00	10.00	3.50	1.06	0.40	8.41	kg.
Second Flight								
Main Bar	1	12.00	7.00	11.00	3.34	0.89	20.81	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
Distribution bar	1	8.00	22.00	3.50	1.06	0.40	9.25	kg.
	1	8.00	10.00	3.50	1.06	0.40	4.20	kg.
	1	8.00	8.00	3.50	1.06	0.40	3.36	kg.
Stirrups								
GriD1-1	1	8.00	124.00	2.66	0.81	0.40	39.61	kg.
Grid2-2	1	8.00	96.00	2.66	0.81	0.40	30.67	kg.
Grid3-3	1	8.00	63.00	2.66	0.81	0.40	20.13	kg.
Grid C-C,D-D,E-E	3	8.00	72.00	2.66	0.81	0.40	69.00	kg.
A-A	1	8.00	93.00	2.66	0.81	0.40	29.71	kg.
B-B	1	8.00	30.00	2.66	0.81	0.40	9.58	kg.
First floor								
Column	12	16.00	8.00	10.00	3.04	1.58	461.18	kg.
Grid C-C,D-D,E-E								kg.
	3	16.00	4.00	25.91	7.88	1.58	149.36	kg.
A-A,3 triangle	1	16.00	4.00	36.00	10.94	1.58	69.18	kg.
B-B	1	16.00	4.00	14.00	4.26	1.58	26.90	kg.
Grid 1-1	1	16.00	4.00	45.00	13.68	1.58	86.47	kg.

Grid 2-2	1	16.00	4.00	36.00	10.94	1.58	69.18	kg.
Grid 3-3	1	16.00	4.00	25.00	7.60	1.58	48.04	kg.
Slab								
Down layer								
A-3	1	8.00	72.00	24.00	7.30	0.40	207.53	kg.
A-E	1	8.00	50.00	36.00	10.94	0.40	216.18	kg.
Top layer								
A-3	1	8.00	36.00	23.00	6.99	0.40	99.44	kg.
A-E	1	8.00	21.00	35.00	10.64	0.40	88.27	kg.
Extra Bar	1	8.00	35.00	2.00	0.61	0.40	8.41	kg.
	1	8.00	35.00	2.00	0.61	0.40	8.41	kg.
	1	8.00	35.00	7.00	2.13	0.40	29.42	kg.
	1	8.00	10.00	7.16	2.18	0.40	8.60	kg.
	1	8.00	14.00	4.41	1.34	0.40	7.41	kg.
	1	8.00	10.00	2.25	0.68	0.40	2.70	kg.
Deduction	1	8.00	14.00	2.25	0.68	0.40	3.78	kg.
	-2	8.00	18.00	7.41	2.25	0.40	-32.04	kg.
	-2	8.00	14.00	9.33	2.84	0.40	-31.37	kg.
Staircase								
First flight & second								
Main Bar	2	12.00	7.00	17.00	5.17	0.89	64.31	kg.
	2	12.00	7.00	7.00	2.13	0.89	26.48	kg.
	2	12.00	7.00	5.00	1.52	0.89	18.92	kg.
	2	12.00	7.00	3.00	0.91	0.89	11.35	kg.
	2	12.00	7.00	3.50	1.06	0.89	13.24	kg.
Distrubution bar	2	8.00	34.00	3.50	1.06	0.40	28.58	kg.
	2	8.00	10.00	3.50	1.06	0.40	8.41	kg.
Second Flight								
Main Bar	1	12.00	7.00	11.00	3.34	0.89	20.81	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
Distrubution bar	1	8.00	22.00	3.50	1.06	0.40	9.25	kg.
	1	8.00	10.00	3.50	1.06	0.40	4.20	kg.

	1	8.00	8.00	3.50	1.06	0.40	3.36	kg.
	2	8.00	10.00	3.50	1.06	0.40	8.41	kg.
Second Flight								
Main Bar	1	12.00	7.00	11.00	3.34	0.89	20.81	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
Distrubution bar	1	8.00	22.00	3.50	1.06	0.40	9.25	kg.
	1	8.00	10.00	3.50	1.06	0.40	4.20	kg.
	1	8.00	8.00	3.50	1.06	0.40	3.36	kg.
Stirupps								
GriD1-1	1	8.00	124.00	2.66	0.81	0.40	39.61	kg.
Grid2-2	1	8.00	96.00	2.66	0.81	0.40	30.67	kg.
Grid3-3	1	8.00	63.00	2.66	0.81	0.40	20.13	kg.
Grid C-C,D-D,E-E	3	8.00	72.00	2.66	0.81	0.40	69.00	kg.
A-A	1	8.00	93.00	2.66	0.81	0.40	29.71	kg.
B-B	1	8.00	30.00	2.66	0.81	0.40	9.58	kg.
Column	12	8.00	18.00	3.50	1.06	0.40	90.79	kg.
inner	12	8.00	18.00	2.55	0.78	0.40	66.15	kg.
Second floor								
Column	12	16.00	4.00	10.00	3.04	1.58	230.59	kg.
A-E	1	8.00	50.00	36.00	10.94	0.40	216.18	kg.
Top layer								kg.
A-3	1	8.00	36.00	23.00	6.99	0.40	99.44	kg.
A-E	1	8.00	21.00	35.00	10.64	0.40	88.27	kg.
Extra Bar	1	8.00	35.00	2.00	0.61	0.40	8.41	kg.
	1	8.00	35.00	2.00	0.61	0.40	8.41	kg.
	1	8.00	35.00	7.00	2.13	0.40	29.42	kg.
	1	8.00	10.00	7.16	2.18	0.40	8.60	kg.
	1	8.00	14.00	4.41	1.34	0.40	7.41	kg.
	1	8.00	10.00	2.25	0.68	0.40	2.70	kg.
Deduction	1	8.00	14.00	2.25	0.68	0.40	3.78	kg.

	-2	8.00	18.00	7.41	2.25	0.40	-32.04	kg.
	-2	8.00	14.00	9.33	2.84	0.40	-31.37	kg.
Staircase								
First flight & second								
Main Bar	2	12.00	7.00	17.00	5.17	0.89	64.31	kg.
	2	12.00	7.00	7.00	2.13	0.89	26.48	kg.
	2	12.00	7.00	5.00	1.52	0.89	18.92	kg.
	2	12.00	7.00	3.00	0.91	0.89	11.35	kg.
	2	12.00	7.00	3.50	1.06	0.89	13.24	kg.
Distribution bar	2	8.00	34.00	3.50	1.06	0.40	28.58	kg.
	2	8.00	10.00	3.50	1.06	0.40	8.41	kg.
Second Flight								
Main Bar	1	12.00	7.00	11.00	3.34	0.89	20.81	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
Distribution bar	1	8.00	22.00	3.50	1.06	0.40	9.25	kg.
	1	8.00	10.00	3.50	1.06	0.40	4.20	kg.
	1	8.00	8.00	3.50	1.06	0.40	3.36	kg.
	2	8.00	10.00	3.50	1.06	0.40	8.41	kg.
Second Flight								
Main Bar	1	12.00	7.00	11.00	3.34	0.89	20.81	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
Distribution bar	1	8.00	22.00	3.50	1.06	0.40	9.25	kg.
	1	8.00	10.00	3.50	1.06	0.40	4.20	kg.
	1	8.00	8.00	3.50	1.06	0.40	3.36	kg.
Stirrups								
GriD1-1	1	8.00	124.00	2.66	0.81	0.40	39.61	kg.
Grid2-2	1	8.00	96.00	2.66	0.81	0.40	30.67	kg.
Grid3-3	1	8.00	63.00	2.66	0.81	0.40	20.13	kg.

Grid C-C,D-D,E-E	3	8.00	72.00	2.66	0.81	0.40	69.00	kg.
A-A	1	8.00	93.00	2.66	0.81	0.40	29.71	kg.
B-B	1	8.00	30.00	2.66	0.81	0.40	9.58	kg.
Column	12	8.00	18.00	3.50	1.06	0.40	90.79	kg.
inner	12	8.00	18.00	2.55	0.78	0.40	66.15	kg.
Third floor								
Column	12	16.00	4.00	10.00	3.04	1.58	230.59	kg.
	12	12.00	4.00	10.00	3.04	0.89	129.71	kg.
Grid C-C,D-D,E-E								
	3	16.00	4.00	25.91	7.88	1.58	149.36	kg.
A-A,3 triangle	1	16.00	4.00	36.00	10.94	1.58	69.18	kg.
B-B	1	16.00	4.00	14.00	4.26	1.58	26.90	kg.
Grid 1-1	1	16.00	4.00	45.00	13.68	1.58	86.47	kg.
Grid 2-2	1	16.00	4.00	36.00	10.94	1.58	69.18	kg.
Grid 3-3	1	16.00	4.00	25.00	7.60	1.58	48.04	kg.
Slab								
Down layer								
A-3	1	8.00	72.00	24.00	7.30	0.40	207.53	kg.
A-E	1	8.00	50.00	36.00	10.94	0.40	216.18	kg.
Top layer								
A-3	1	8.00	36.00	23.00	6.99	0.40	99.44	kg.
A-E	1	8.00	21.00	35.00	10.64	0.40	88.27	kg.
Extra Bar	1	8.00	35.00	2.00	0.61	0.40	8.41	kg.
	1	8.00	35.00	2.00	0.61	0.40	8.41	kg.
	1	8.00	35.00	7.00	2.13	0.40	29.42	kg.
	1	8.00	10.00	7.16	2.18	0.40	8.60	kg.
	1	8.00	14.00	4.41	1.34	0.40	7.41	kg.
	1	8.00	10.00	2.25	0.68	0.40	2.70	kg.
Deduction	1	8.00	14.00	2.25	0.68	0.40	3.78	kg.
	-2	8.00	18.00	7.41	2.25	0.40	-32.04	kg.
	-2	8.00	14.00	9.33	2.84	0.40	-31.37	kg.
Staircase								
First flight & second								
Main Bar	2	12.00	7.00	17.00	5.17	0.89	64.31	kg.
	2	12.00	7.00	7.00	2.13	0.89	26.48	kg.

	2	12.00	7.00	5.00	1.52	0.89	18.92	kg.
	2	12.00	7.00	3.00	0.91	0.89	11.35	kg.
	2	12.00	7.00	3.50	1.06	0.89	13.24	kg.
Distribution bar	2	8.00	34.00	3.50	1.06	0.40	28.58	kg.
	2	8.00	10.00	3.50	1.06	0.40	8.41	kg.
Second Flight								
Main Bar	1	12.00	7.00	11.00	3.34	0.89	20.81	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
Distribution bar	1	8.00	22.00	3.50	1.06	0.40	9.25	kg.
	1	8.00	10.00	3.50	1.06	0.40	4.20	kg.
	1	8.00	8.00	3.50	1.06	0.40	3.36	kg.
	2	8.00	10.00	3.50	1.06	0.40	8.41	kg.
Second Flight								
Main Bar	1	12.00	7.00	11.00	3.34	0.89	20.81	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	5.00	1.52	0.89	9.46	kg.
	1	12.00	7.00	4.00	1.22	0.89	7.57	kg.
Distribution bar	1	8.00	22.00	3.50	1.06	0.40	9.25	kg.
	1	8.00	10.00	3.50	1.06	0.40	4.20	kg.
	1	8.00	8.00	3.50	1.06	0.40	3.36	kg.
Stirrups								
GriD1-1	1	8.00	124.00	2.66	0.81	0.40	39.61	kg.
Grid2-2	1	8.00	96.00	2.66	0.81	0.40	30.67	kg.
Grid3-3	1	8.00	63.00	2.66	0.81	0.40	20.13	kg.
Grid C-C,D-D,E-E	3	8.00	72.00	2.66	0.81	0.40	69.00	kg.
A-A	1	8.00	93.00	2.66	0.81	0.40	29.71	kg.
B-B	1	8.00	30.00	2.66	0.81	0.40	9.58	kg.
Column	12	8.00	18.00	3.50	1.06	0.40	90.79	kg.
inner	12	8.00	18.00	2.55	0.78	0.40	66.15	kg.

Fourth Floor								
Slab	2	8.00	26.00	24.00	7.30	0.40	149.88	kg.
	2	8.00	50.00	13.00	3.95	0.40	156.13	kg.
Column	6	16.00	4.00	10.00	3.04	1.58	115.29	kg.
	6	12.00	4.00	10.00	3.04	0.89	64.85	kg.
Beam								
	3	16.00	4.00	18.00	5.47	1.58	103.77	kg.
	2	16.00	4.00	26.00	7.90	1.58	99.92	kg.
Stirrups								
Beam	3	8.00	26.00	2.66	0.81	0.40	24.92	kg.
	2	8.00	50.00	2.66	0.81	0.40	31.95	kg.
Column	6	8.00	20.00	3.16	0.96	0.40	45.54	kg.
Inner	6	8.00	20.00	2.66	0.81	0.40	38.34	kg.
							15500.00	kg.

Table 8 : Reinforcement Work

FORMWORK:

Description of work	No	Length	Breadth	Height	Quantity	Unit
Rectangle portion						
<i>F1</i>	1	34.00	0.75		25.50	sq. ft.
<i>F2</i>	2	30.00	0.75		45.00	sq. ft.
<i>F3</i>	3	25.00	0.75		56.25	sq. ft.
<i>F4</i>	3	24.00	0.75		54.00	sq. ft.
<i>F5</i>	3	29.00	0.75		65.25	sq. ft.
Plinth level Beam						
,C-C,D-D,E-E	4	20.91	2.00		167.28	sq. ft.
A-A	1	29.83	2.00		59.66	sq. ft.
B-B	1	9.25	2.00		18.50	sq. ft.
GriD1-1	1	36.41	2.00		72.82	sq. ft.
Grid2-2	1	28.00	2.00		56.00	sq. ft.
Grid3-3	1	17.83	2.00		35.66	sq. ft.
Pedestal Column	12	4.00	3.66		175.68	sq. ft.
Ground floor						
Column	12	4.00	9.00		432.00	sq. ft.
Beam						
,C-C,D-D,E-E	4	20.91	2.75		230.01	sq. ft.
A-A	1	29.83	2.75		82.03	sq. ft.
B-B	1	9.25	2.75		25.44	sq. ft.
GriD1-1	1	36.41	2.75		100.13	sq. ft.
Grid2-2	1	28.00	2.75		77.00	sq. ft.
Grid3-3	1	17.83	2.75		49.03	sq. ft.
Slab	1	1.00	745.00		745.00	sq. ft.
Staircase						
First Floor & Third Flight	2	5.50	3.50		38.50	sq. ft.
Second Flight	1	2.50	3.50		8.75	sq. ft.
First & Third step	10	3.50	1.00		35.00	sq. ft.
Second Step	2	3.50	1.00		7.00	sq. ft.
Landing						
First & Second	2	3.50	3.50		24.50	sq. ft.
Deduction part	-1	8.33	8.33		-69.39	sq. ft.

First floor						
Column	12	4.00	7.25	7.25	348.00	sq. ft.
Beam						
,C-C,D-D,E-E	4	20.91	2.75		230.01	sq. ft.
A-A	1	29.83	2.75		82.03	sq. ft.
B-B	1	9.25	2.75		25.44	sq. ft.
GriD1-1	1	36.41	2.75		100.13	sq. ft.
Grid2-2	1	28.00	2.75		77.00	sq. ft.
Grid3-3	1	17.83	2.75		49.03	sq. ft.
Slab	1	1.00	826.00		826.00	sq. ft.
Staircase						
First Floor & Third Flight	2	5.50	3.50		38.50	sq. ft.
second flight	1	2.50	3.50		8.75	sq. ft.
First & Third step	10	3.50	1.00		35.00	sq. ft.
Second Step	2	3.50	1.00		7.00	sq. ft.
Landing						sq. ft.
First & Second	2	3.50	3.50		24.50	sq. ft.
Deduction part	-1	8.33	8.33		-69.39	sq. ft.
Second floor						
Column	12	4.00	7.25		348.00	sq. ft.
Beam						
,C-C,D-D,E-E	4	20.91	2.75		230.01	sq. ft.
A-A	1	29.83	2.75		82.03	sq. ft.
B-B	1	9.25	2.75		25.44	sq. ft.
GriD1-1	1	36.41	2.75		100.13	sq. ft.
Grid2-2	1	28.00	2.75		77.00	sq. ft.
Grid3-3	1	17.83	2.75		49.03	sq. ft.
Slab	1	1.00	826.00		826.00	sq. ft.
Staircase						
First Floor & Third Flight	2	5.50	3.50		38.50	sq. ft.
second flight	1	2.50	3.50		8.75	sq. ft.
First & Third step	10	3.50	1.00		35.00	sq. ft.
Second Step	2	3.50	1.00		7.00	sq. ft.
Landing						
First & Second	2	3.50	3.50		24.50	sq. ft.
Deduction part	-1	8.33	8.33		-69.39	sq. ft.

Third floor						
Column	12	4.00	7.25		348.00	sq. ft.
Beam						
,C-C,D-D,E-E	4	20.91	2.75		230.01	sq. ft.
A-A	1	29.83	2.75		82.03	sq. ft.
B-B	1	9.25	2.75		25.44	sq. ft.
GriD1-1	1	36.41	2.75		100.13	sq. ft.
Grid2-2	1	28.00	2.75		77.00	sq. ft.
Grid3-3	1	17.83	2.75		49.03	sq. ft.
Slab	1	1.00	826.00		826.00	sq. ft.
Staircase						
First Floor & Third Flight	2	5.50	3.50		38.50	sq. ft.
Second Flight	1	2.50	3.50		8.75	sq. ft.
First & Third Step	10	3.50	1.00		35.00	sq. ft.
Second Step	2	3.50	1.00		7.00	sq. ft.
Landing						
First & Second	2	3.50	3.50		24.50	sq. ft.
Deduction part	-1	8.33	8.33		-69.39	sq. ft.
Fourth floor						
Slab	1	1.00	327.00		327.00	sq. ft.
Top	1	45.00	1.50		67.50	sq. ft.
Top	1	30.00	1.50		45.00	sq. ft.
Beam	2	24.00	2.75		132.00	sq. ft.
	3	13.00	2.75		107.25	sq. ft.
Column	6	4.00	2.75		66.00	sq. ft.
		Sub total			10500.00	sq. ft.

Table 9 : Formwork

BRICKWORK:

Description of work	No	Length	Breadth	Height	Quantity	Unit
1) Brick work (9")						
Grid 1-1	1	35.00	1.50	1.00	52.50	cu. ft.
Grid 1-1	1	35.00	1.16	1.00	40.60	cu. ft.
Grid 1-1	1	35.00	0.75	1.16	30.45	cu. ft.
Grid 2-2	1	28.00	1.50	1.00	42.00	cu. ft.
Grid 2-2	1	28.00	1.16	1.00	32.48	cu. ft.
Grid 2-2	1	28.00	0.75	1.16	24.36	cu. ft.
Grid 3-3	1	17.83	1.50	1.00	26.75	cu. ft.
Grid 3-3	1	17.83	1.16	1.00	20.68	cu. ft.
Grid 3-3	1	17.83	0.75	1.16	15.51	cu. ft.
Grid c-c,D-D,E-E	3	20.91	1.50	1.00	94.10	cu. ft.
Grid c-c,D-D,E-E	3	20.91	1.16	1.00	72.77	cu. ft.

Grid c-c,D-D,E-E	3	20.91	0.75	1.16	54.58	cu. ft.
A- 3	1	29.00	1.50	1.00	43.50	cu. ft.
A- 3	1	29.00	1.16	1.00	33.64	cu. ft.
A- 3	1	29.00	0.75	1.16	25.23	cu. ft.
B-B	1	9.25	1.50	1.00	13.88	cu. ft.
	1	9.25	1.16	1.00	10.73	cu. ft.
	1	9.25	0.75	1.16	8.05	cu. ft.
Ground floor						
Grid E-E	1	21.00	0.75	9.00	15.75	cu. ft.
Grid 3-3	1	17.83	0.75	9.00	13.37	cu. ft.
Triangle	1	11.00	0.75	9.00	8.25	cu. ft.
MD	-1	3.50	0.75	8.25	-2.63	cu. ft.
W4	-1	3.25	0.75	4.00	-2.44	cu. ft.
First floor						
Grid 2-2	1	3.66	0.75	9.00	2.75	cu. ft.
Grid 3-3	1	21.49	0.75	9.00	16.12	cu. ft.
D	-1	3.25	0.75	4.00	-2.44	cu. ft.
W4	-1	3.25	0.75	4.00	-2.44	cu. ft.
Second floor	1	21.40	0.75	8.00	16.05	cu. ft.
Grid E-E	1	12.00	0.75	8.00	9.00	cu. ft.
D	-1	3.25	0.75	4.00	-2.44	cu. ft.
W4	-1	3.25	0.75	4.00	-2.44	cu. ft.
Third floor	1	21.40	0.75	8.00	16.05	cu. ft.
D	-1	3.25	0.75	4.00	-2.44	cu. ft.
W4	-1	3.25	0.75	4.00	-2.44	cu. ft.
Forth floor						
Grid 3-3	1	16.00	0.75	8.00	12.00	cu. ft.
Grid E-E	1	12.00	0.75	8.00	9.00	cu. ft.
Grid D-D	1	12.00	0.75	8.00	9.00	cu. ft.
D	-2	3.25	0.75	4.00	-4.88	cu. ft.
W4	-1	3.25	0.75	4.00	-2.44	cu. ft.
		Subtotal			1081.00	cu. ft.

2) Brick work (4")						
Ground floor						
Grid 2-2	1	28.00		7.75	217.00	cu. ft.
Grid c-c	1	9.25		7.75	71.69	cu. ft.
Grid D-D	1	20.83		7.75	161.43	cu. ft.
D	-1	3.25		4.00	-13.00	cu. ft.

Void cover	1	6.50		7.75	50.38	cu. ft.
First floor						
Grid 1-1	1	35.00		6.75	236.25	cu. ft.
Grid 2-2	1	28.00		6.75	189.00	cu. ft.
Grid A-A	1	29.00		6.75	195.75	cu. ft.
C-C,,E-E	2	9.25		6.75	124.88	cu. ft.
D-D	1	20.19		6.75	136.28	cu. ft.
D	-3	3.25		4.00	-39.00	cu. ft.
MD	1	3.25		8.25	26.81	cu. ft.
D1	-1	2.50		6.50	-16.25	cu. ft.
Bathroom Wall	1	24.00		6.75	162.00	cu. ft.
Parapet Wall	1	11.58		1.83	21.19	cu. ft.
w1	-4	7.00		6.00	-168.00	cu. ft.
w2	-2	8.00		6.00	-96.00	cu. ft.
w3	-1	6.00		6.00	-36.00	cu. ft.
Second floor						
Grid 1-1	1	35.00		6.75	236.25	cu. ft.
Grid 2-2	1	28.00		6.75	189.00	cu. ft.
Grid A-A	1	29.00		6.75	195.75	cu. ft.
C-C,,E-E	2	9.25		6.75	124.88	cu. ft.
D-D	1	20.19		6.75	136.28	cu. ft.
D	-3	3.25		4.00	-39.00	cu. ft.
MD	1	3.25		8.25	26.81	cu. ft.
D1	-1	2.50		6.50	-16.25	cu. ft.
Bathroom Wall	1	24.00		6.75	162.00	cu. ft.
Parapet Wall	1	11.58		1.83	21.19	cu. ft.
w1	-4	7.00		6.00	-168.00	cu. ft.
w2	-2	8.00		6.00	-96.00	cu. ft.

w3	-1	6.00		6.00	-36.00	cu. ft.
Third floor						
Grid 1-1	1	35.00		6.75	236.25	cu. ft.
Grid 2-2	1	28.00		6.75	189.00	cu. ft.
Grid A-A	1	29.00		6.75	195.75	cu. ft.
C-C,,E-E	2	9.25		6.75	124.88	cu. ft.
D-D	1	20.19		6.75	136.28	cu. ft.
D	-3	3.25		4.00	-39.00	cu. ft.
MD	1	3.25		8.25	26.81	cu. ft.
D1	-1	2.50		6.50	-16.25	cu. ft.
Bathroom Wall	1	24.00		6.75	162.00	cu. ft.
Parapet Wall	1	11.58		1.83	21.19	cu. ft.
w1	-4	7.00		6.00	-168.00	cu. ft.
w2	-2	8.00		6.00	-96.00	cu. ft.
w3	-1	6.00		6.00	-36.00	cu. ft.
Forth floor						
Grid 1-1	1	13.00		6.75	87.75	cu. ft.
Grid 2-2	1	8.08		6.75	54.54	cu. ft.
Grid E-E	1	14.00		6.75	94.50	cu. ft.
Grid D-D	1	9.25		6.75	62.44	cu. ft.
Balcony Parapet wall	1	11.58		1.83	21.19	cu. ft.
Parapet wall Fourth floor	1	68.41		3.00	205.23	cu. ft.
Top floor	1					
D	-2	3.25	0.75	6.75	-43.88	cu. ft.
W4	-1	3.25	0.75	6.75	-2.44	cu. ft.
		subtotal			3477.00	cu. ft.
Face wall						

North side	2	1.00	22.00		44.00	cu. ft.
South side	1	1.00	220.00		220.00	cu. ft.
East side	2	1.00	19.00		38.00	cu. ft.
West side	1	1.00	32.00		32.00	cu. ft.
Balcony	4	1.00	22.00		88.00	cu. ft.
		subtotal			522.00	cu. ft.

Table 10 : Brickwork

ALUMINIUM WORK:

Description of work	No	Length	Breadth	Height	Quantity	Unit
MD1	3	3.25	8.25		80.44	sq. ft
MD	1	3.50	8.25		28.88	sq. ft
w1	13	7.00	6.00		546.00	sq. ft
w2	6	8.00	6.00		288.00	sq. ft
w3	5	6.00	6.00		180.00	sq. ft
w4	5	3.25	4.00		65.00	sq. ft
		Sub total			1188.31	sq. ft

Table 11 : Aluminum Work

TILE WORK:

Description of work	No	Length	Breadth	Height	Quantity	Unit
Wall Tile						
First floor	1	22.00	6.00		132.00	sq. ft.
Second and Third	2	22.00	6.00		264.00	sq. ft.
Floor Tile	3	6.00	4.00		72.00	sq. ft.
Kitchen slab	3	1.00	21.00		63.00	sq. ft.
Kausi	1	420.00	1.00		420.00	sq. ft.
		Sub total			1100.00	sq. ft.

Table 12 : Tile Work

LINTEL WORK:

Description of work	No	Length	Breadth	Height	Quantity	Unit
1) Brick work (9")						
Ground floor						
Grid E-E	1	21.00		1.00	21.00	sq. ft.
Grid 3-3	1	17.83		1.00	17.83	sq. ft.
Triangle	1	11.00		1.00	11.00	sq. ft.
First floor				1.00		
Grid 2-2	1	3.66		1.00	3.66	sq. ft.
Grid 3-3	1	21.49		1.00	21.49	sq. ft.
Second floor	1	21.40		1.00	21.40	sq. ft.
Grid E-E	1	12.00		1.00	12.00	sq. ft.
Third floor	1	21.40		1.00	21.40	sq. ft.
Forth floor						
Grid 3-3	1	16.00		1.00	16.00	sq. ft.
Grid E-E	1	12.00		1.00	12.00	sq. ft.
Grid D-D	1	12.00		1.00	12.00	sq. ft.
2) Brick work (4")						
Ground floor						
Grid 2-2	1	28.00		1.00	28.00	sq. ft.
Grid c-c	1	9.25		1.00	9.25	sq. ft.
Grid D-D	1	20.83		1.00	20.83	sq. ft.
Void cover	1	6.50		1.00	6.50	sq. ft.
First floor						
Grid 1-1	1	35.00		1.00	35.00	sq. ft.
Grid 2-2	1	28.00		1.00	28.00	sq. ft.
Grid A-A	1	29.00		1.00	29.00	sq. ft.
C-C,,E-E	2	9.25		1.00	18.50	sq. ft.
D-D	1	20.19		1.00	20.19	sq. ft.
MD	1	3.25		1.00	3.25	sq. ft.
Bathroom Wall	1	24.00		1.00	24.00	sq. ft.
Second floor	1	21.40		1.00	21.40	sq. ft.
Grid 1-1	1	35.00		1.00	35.00	sq. ft.
Grid 2-2	1	28.00		1.00	28.00	sq. ft.
Grid A-A	1	29.00		1.00	29.00	sq. ft.
C-C,,E-E	2	9.25		1.00	18.50	sq. ft.
D-D	1	20.19		1.00	20.19	sq. ft.
MD	1	3.25		1.00	3.25	sq. ft.
Bathroom Wall	1	24.00		1.00	24.00	sq. ft.
Third floor						
Grid 1-1	1	35.00		1.00	35.00	sq. ft.
Grid 2-2	1	28.00		1.00	28.00	sq. ft.

Grid A-A	1	29.00		1.00	29.00	sq. ft.
C-C,,E-E	2	9.25		1.00	18.50	sq. ft.
D-D	1	20.19		1.00	20.19	sq. ft.
Bathroom Wall	1	24.00		1.00	24.00	sq. ft.
Fourth floor						
Grid 1-1	1	13.00		1.00	13.00	sq. ft.
Grid 2-2	1	8.08		1.00	8.08	sq. ft.
Grid E-E	1	14.00		1.00	14.00	sq. ft.
Grid D-D	1	9.25		1.00	9.25	sq. ft.
		subtotal			770.66	sq. ft.

Table 13 : Lintel Work

PLASTER WORK:

Description of work	No	Length	Breadth	Height	Quantity	Unit
Outer						
North side	1	23.00	36.00		828.00	sq. ft.
South side	1	36.00	24.00		864.00	sq. ft.
East side	1	20.00	28.00		560.00	sq. ft.
West side	1	34.00	44.00		1496.00	sq. ft.
Deduction						
North side	-6	6.66	6.00			sq. ft.
South side w4	-4	4.00	3.25			sq. ft.
D1	-4	3.00	7.00			sq. ft.
	-4	6.66	6.00			sq. ft.
East side						
	-6	6.66	6.00			sq. ft.
West side	-13	6.00	7.66			sq. ft.
		Sub total			3748.00	sq. ft.
Inner						sq. ft.
North side	1	23.00	36.00		828.00	sq. ft.
South side	1	36.00	24.00		864.00	sq. ft.
East side	1	20.00	28.00		560.00	sq. ft.

West side	1	34.00	44.00		1496.00	sq. ft.
Deduction						
North side	-6	6.66	6.00		-239.76	sq. ft.
South side w4	-4	4.00	3.25		-52.00	sq. ft.
D1	-4	3.00	7.00		-84.00	sq. ft.
	-4	6.66	6.00		-159.84	sq. ft.
East side						
	-6	6.66	6.00		-239.76	sq. ft.
West side	-13	6.00	7.66		-597.48	sq. ft.
		Sub total			2375.16	sq. ft.
Brick work (4")						
Ground floor						
Grid 2-2	2	28.00		9.00	504.00	sq. ft.
Grid c-c	2	9.25		9.00	166.50	sq. ft.
Grid D-D	2	20.83		9.00	374.94	sq. ft.
D	-1	3.25		7.00	-22.75	sq. ft.
Void cover	1	6.50		9.00	58.50	sq. ft.
First floor						
Grid 2-2	2	28.00		9.00	504.00	sq. ft.
C-C,,	2	9.25		9.00	166.50	sq. ft.
D-D	2	20.19		9.00	363.42	sq. ft.
D	-3	3.25		7.00	-68.25	sq. ft.
MD	2	3.25		8.25	53.63	sq. ft.
D1	2	2.50		6.50	32.50	sq. ft.
Bathroom Wall	2	24.00		9.00	432.00	sq. ft.
parapet wall	1	11.58		1.83	21.19	sq. ft.

Second floor						
Grid 2-2	2	28.00		9.00	504.00	sq. ft.
C-C,,	2	9.25		9.00	166.50	sq. ft.
D-D	2	20.19		9.00	363.42	sq. ft.
D	-3	3.25		7.00	-68.25	sq. ft.
MD	2	3.25		8.25	53.63	sq. ft.
D1	2	2.50		6.50	32.50	sq. ft.
Bathroom Wall	2	24.00		9.00	432.00	sq. ft.
parapet wall	1	11.58		1.83	21.19	sq. ft.
Third floor						
Grid 2-2	2	28.00		9.00	504.00	sq. ft.
C-C,,	2	9.25		9.00	166.50	sq. ft.
D-D	2	20.19		9.00	363.42	sq. ft.
D	-3	3.25		7.00	-68.25	sq. ft.
MD	2	3.25		8.25	53.63	sq. ft.
D1	2	2.50		6.50	32.50	sq. ft.
Bathroom Wall	2	24.00		9.00	432.00	sq. ft.
parapet wall	1	11.58		1.83	21.19	sq. ft.
Bathroom Wall	1	24.00		9.00	216.00	sq. ft.
Fourth floor						
Grid 2-2	2	8.08		9.00	145.44	sq. ft.
		subtotal			10100.00	sq. ft.

Table 14 : Plaster Work

PAINTING WORK:

Description of work	No	Length	Breadth	Height	Quantity	Unit
Weather Coat						
North side	1	23.00	36.00		828.00	sq. ft.
South side	1	36.00	24.00		864.00	sq. ft.
East side	1	20.00	28.00		560.00	sq. ft.
West side	1	34.00	44.00		1496.00	sq. ft.
Deduction						
North side	-6	6.66	6.00		-239.76	sq. ft.
South sidew4	-4	4.00	3.25		-52.00	sq. ft.
D1	-4	3.00	7.00		-84.00	sq. ft.
	-4	6.66	6.00		-159.84	sq. ft.

East side						
	-6	6.66	6.00		-239.76	sq. ft.
West side	-13	6.00	7.66		-597.48	sq. ft.
		Sub total			2375.16	sq. ft.
Distemper						
North side	1	23.00	36.00		828.00	sq. ft.
South side	1	36.00	24.00		864.00	sq. ft.
East side	1	20.00	28.00		560.00	sq. ft.
West side	1	34.00	44.00		1496.00	sq. ft.
Deduction						
North side	-6	6.66	6.00		-239.76	sq. ft.
South sidew4	-4	4.00	3.25		-52.00	sq. ft.
D1	-4	3.00	7.00		-84.00	sq. ft.
	-4	6.66	6.00		-159.84	sq. ft.
East side						
	-6	6.66	6.00		-239.76	sq. ft.
West side	-13	6.00	7.66		-597.48	sq. ft.
		Sub total			2375.16	sq. ft.
Brick work (4")						
Ground floor						
Grid 2-2	2	28.00		9.00	504.00	sq. ft.
Grid c-c	2	9.25		9.00	166.50	sq. ft.
Grid D-D	2	20.83		9.00	374.94	sq. ft.
D	-1	3.25		7.00	-22.75	sq. ft.
Void cover	1	6.50		9.00	58.50	sq. ft.

First floor						
Grid 2-2	2	28.00		9.00	504.00	sq. ft.
C-C,,	2	9.25		9.00	166.50	sq. ft.
D-D	2	20.19		9.00	363.42	sq. ft.
D	-3	3.25		7.00	-68.25	sq. ft.
MD	2	3.25		8.25	53.63	sq. ft.
D1	2	2.50		6.50	32.50	sq. ft.
Bathroom Wall	2	24.00		9.00	432.00	sq. ft.
parapet wall	1	11.58		1.83	21.19	sq. ft.
Second floor						
Grid 2-2	2	28.00		9.00	504.00	sq. ft.
C-C,,	2	9.25		9.00	166.50	sq. ft.
D-D	2	20.19		9.00	363.42	sq. ft.
D	-3	3.25		7.00	-68.25	sq. ft.
MD	2	3.25		8.25	53.63	sq. ft.
D1	2	2.50		6.50	32.50	sq. ft.
Bathroom Wall	2	24.00		9.00	432.00	sq. ft.
parapet wall	1	11.58		1.83	21.19	sq. ft.
Third floor						
Grid 2-2	2	28.00		9.00	504.00	sq. ft.
C-C,,	2	9.25		9.00	166.50	sq. ft.
D-D	2	20.19		9.00	363.42	sq. ft.
D	-3	3.25		7.00	-68.25	sq. ft.
MD	2	3.25		8.25	53.63	sq. ft.
D1	2	2.50		6.50	32.50	sq. ft.
Bathroom Wall	2	24.00		9.00	432.00	sq. ft.
parapet wall	1	11.58		1.83	21.19	sq. ft.
Bathroom Wall	1	24.00		9.00	216.00	sq. ft.
Forth floor						
Grid 2-2	2	8.08		9.00	145.44	sq. ft.
		subtotal			10100.00	sq. ft.

Table 15 : Painting Work

KORA MASINO WORK:

Description of work	No	Length	Breadth	Height	Quantity	Unit
Koramasino		Area				
Ground floor	1	756.00	1.00		756.00	sq. ft.
First floor	1	836.00	1.00		836.00	sq. ft.
Second floor	1	837.00	1.00		837.00	sq. ft.
Third floor	1	838.00	1.00		838.00	sq. ft.
Fourth floor	1	130.00	1.00		130.00	sq. ft.
	1	36.00	1.00		36.00	sq. ft.
	1	40.00	1.00		40.00	sq. ft.
	1	468.00	1.00		468.00	sq. ft.
Top floor	1	300.00	1.00		300.00	sq. ft.
		Sub total				sq. ft.
Deduction						
Bathroom	-3	6.00	5.00		-90.00	sq. ft.
Staircase	-4	9.33	7.66		-285.87	sq. ft.
			Sub total		3965.00	sq. ft.

Table 16 : Kora Masino Work

STAINLESS STEEL:

Description of work	No	Length	Breadth	Height	Quantity
Stainless stell					
South side	5	11.75	1.00		58.75
Staircase	4	10.00	1.00		40.00
		Sub total			155.00

Table 17 : Stainless Steel

MARBLE WORK:

Description of work	No	Length	Breadth	Height	Quantity	Unit
Marble		Area				
Ground floor						
Riser	15	3.50	1.00		52.50	sq. ft.
Tread	12	3.50	1.00		42.00	sq. ft.
Landing	1	7.00	7.00		49.00	sq. ft.
First floor						
Riser	15	3.50	1.00		52.50	sq. ft.
Tread	12	3.50	1.00		42.00	sq. ft.
Landing	1	7.00	7.00		49.00	sq. ft.
Second floor						
Riser	15	3.50	1.00		52.50	sq. ft.
Tread	12	3.50	1.00		42.00	sq. ft.
Landing	1	7.00	7.00		49.00	sq. ft.
Third floor	1					
Riser	15	3.50	1.00		52.50	sq. ft.
Tread	12	3.50	1.00		42.00	sq. ft.
Landing	1	7.00	7.00		49.00	sq. ft.
		Sub total			1765.00	sq. ft.

Table 18 : Marble Work

DOOR AND WINDOWS:

Description of work	No	Length	Breadth	Height	Quantity	Unit
Door		Area				
D	15	21.00	0.25	0.33	25.99	cu. ft.
D1	3	15.50	0.25	0.33	3.84	cu. ft.
MD1	3	19.75	0.25	0.33	4.89	cu. ft.
MD	1	20.00	0.25	0.33	1.65	cu. ft.
w1	13	26.00	0.25	0.33	27.89	cu. ft.
w2	6	28.00	0.25	0.33	13.86	cu. ft.
for Akhi Jhyal	29	6.50	0.25	0.33	15.55	cu. ft.
w3	5	24.00	0.25	0.33	9.90	cu. ft.
w4	5	14.50	0.25	0.33	5.98	cu. ft.
Rolling Shutter		Sub total			109.54	cu. ft.
RL	1	8.33	8.25		68.72	cu. ft.
RL1	1	9.50	8.25		78.38	cu. ft.
RL2	1	9.75	8.25		80.44	cu. ft.
RL3	1	5.75	8.25		47.44	cu. ft.
RL4	1	9.83	8.25		81.10	cu. ft.
RL5	1	12.58	21.00		264.18	cu. ft.
		Sub total			620.25	cu. ft.
Khapa		Area				
D	15	3.00	7.50		337.50	cu. ft.
D1	3	2.50	6.50		48.75	cu. ft.
MD1	3	3.25	8.25		80.44	cu. ft.
		Sub total			466.69	cu. ft.
		Total			1196.48	cu. ft.

Table 19 : Doors and Windows

SUMMARY OF CALCULATIONS

The summary of the entire estimation including the quantities and rate is as shown in table below:

S.N.	Description of work	Unit	Rate	Quantity	Total Amount
1	Site clearance (Before & After)		50000	1.00	50000.00
2	Soil filling	cu. ft.	15	3397.50	50962.50
3	Excavation of work	cu. ft.	20	3071.55	61431.01
4	Soling	sq. ft.	70	1359.31	95151.71
5	PCC	cu. ft.	390	339.83	132532.73
6	Brick work 9"	cu. ft.	425	1081.00	459425.00
7	Brick work 4"	sq. ft.	175	3477.00	608475.00
8	RCC work	cu. ft.	475	3985.00	1892875.00
9	Reinforcement	kg	112	15500.00	1736000.00
10	Formwork	sq. ft.	80	10500.00	840000.00
11	Plaster (inner)	sq. ft.	55	10100.00	555500.00
12	Plaster (outer)	sq. ft.	60	3748.00	224880.00
13	Weather Coat	sq. ft.	22	2375.16	52253.52
14	Distemper	sq. ft.	16	10100.00	161600.00
15	Shutter	sq. ft.	300	620.25	186075.00
16	Tile	sq. ft.	280	1100.00	308000.00
17	Marble	sq. ft.	420	1765.00	741300.00
18	Aluminium	sq. ft.	620	1405.81	871603.75
19	Door & windows Frame	cu. ft.	7500	109.54	821545.31
20	Stainless Steel for Staircase & Railing	R. ft	1300	155.00	201500.00
21	Lintel	R. ft	420	770.66	323677.20
22	Water Proofing Polymer Coating	sq. ft.	42	291.00	12222.00
	Total				10387009.73
	3% Contingencies and 2% Work Charge Establishment				519350.49
	8% Electricity and 6% Plumbing Works				1454181.36
	Grand Total				12360541.58

Table 20 : Abstract of Cost

PLINTH AREA RATE:

Total Area	3874.00	sq. ft.
Plinth Area Rate	3190.64	Nrs./sq. ft.

F. CONCLUSION

Based on the detailed estimation and valuation of the residential building of four storeyed, it has been determined that the total cost of the project is about Rs. One Crore Twenty Three Lakh. The plinth area rate for the building was determined to be about Rs. 3200/sq. ft. These findings suggest that the construction of the residential building has required a significant financial investment, but the final product has likely to be of high quality and have a strong market value.

